H2 Math Summary

Notes

Maximus Pung

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1 Systems of Linear Equations

1.1 Forming the Equations

We read the context of the question and form equations depending on the context of the question.

Example 1.1

$$\left.\begin{aligned} a+b+c&=176 && —(1)\\ 0.85a+0.90b+0.95c&=161.80 && —(2)\\ 0.93a+0.95b+0.98c&=169.56 && —(3)\end{aligned}\right\} \Rightarrow a=32,\ b=44,\ c=100$$

Example 1.2

The "variables" may be expressions. To find x, y and z, use the given information to form the equations:

$$\left.\begin{aligned} 2x^3+3\ln y-\mathrm{e}^z&=18 && —(1)\\ x^3-\ln y+2\mathrm{e}^z&=9 && —(2)\\ x^3+2\ln y+\mathrm{e}^z&=11 && —(3)\end{aligned}\right\} \Rightarrow x^3=8,\ \ln y=1,\ \mathrm{e}^z=1$$

It is technically still 'linear' because we can think of $A=x^3$, $B=\ln y$, $C=\mathrm{e}^z$, which turns it into a linear system in A, B and C. A truly nonlinear equation could look like $x^2+x+y^2+2y-3z^3-xz+xy=1$; it cannot be made linear this way, because x appears as both x^2 and x, and there are product terms such as xz and xy.

1.2 Special Case: Number of Equations > Number of Variables

Sometimes the number of equations is more than the number of variables, and the solution must satisfy all of the equations.

Example 1.3

$$\left.\begin{aligned} \lambda+\mu&=1 && —(1)\\ 2\lambda-\mu&=5 && —(2)\end{aligned}\right\} \Rightarrow \lambda=2,\ \mu=-1$$

$$\lambda+3\mu=-1 \qquad —(3)$$

Sometimes there may be no solution, e.g. skew lines in vectors. In this case, solve for λ and μ using equations (1) and (2). Then sub into (3).

> **Note**
>
> Check (3): if LHS = RHS, the equations are consistent; if LHS ≠ RHS, there is no solution.

1.3 Special Case: Number of Equations < Number of Variables

Sometimes the number of equations is less than the number of variables, so there are infinitely many solutions.

Example 1.4

$$\left.\begin{aligned} x+y+z=6 \quad &—(1) \\ x-y+2z=5 \quad &—(2) \end{aligned}\right\} \Rightarrow x=\tfrac{11}{2}-\tfrac{3}{2}z,\ y=\tfrac{1}{2}+\tfrac{1}{2}z,\ z=z$$

> **Note**
>
> Useful trick: in contextual questions, the variables are usually non-negative integers, $z \in \mathbb{Z}_0^+$, so use this to find the solution from the general solution.

When $z=z$, it means that there are infinitely many solutions, and z can be any real value. So just sub that into x and y to get a correct solution.

$z=-1: (7,0,-1)$ $z=0: (\tfrac{11}{2},\tfrac{1}{2},0)$ $z=1: (4,1,1)$ $z=2: (\tfrac{5}{2},\tfrac{3}{2},2)$

$z=3: (1,2,3)$ $z=4: (-\tfrac{1}{2},\tfrac{5}{2},4)$ $z=5: (-2,3,5)$ $z=6: (-\tfrac{7}{2},\tfrac{7}{2},6)$

We can use the GC table to find all these solutions.

2 Inequalities

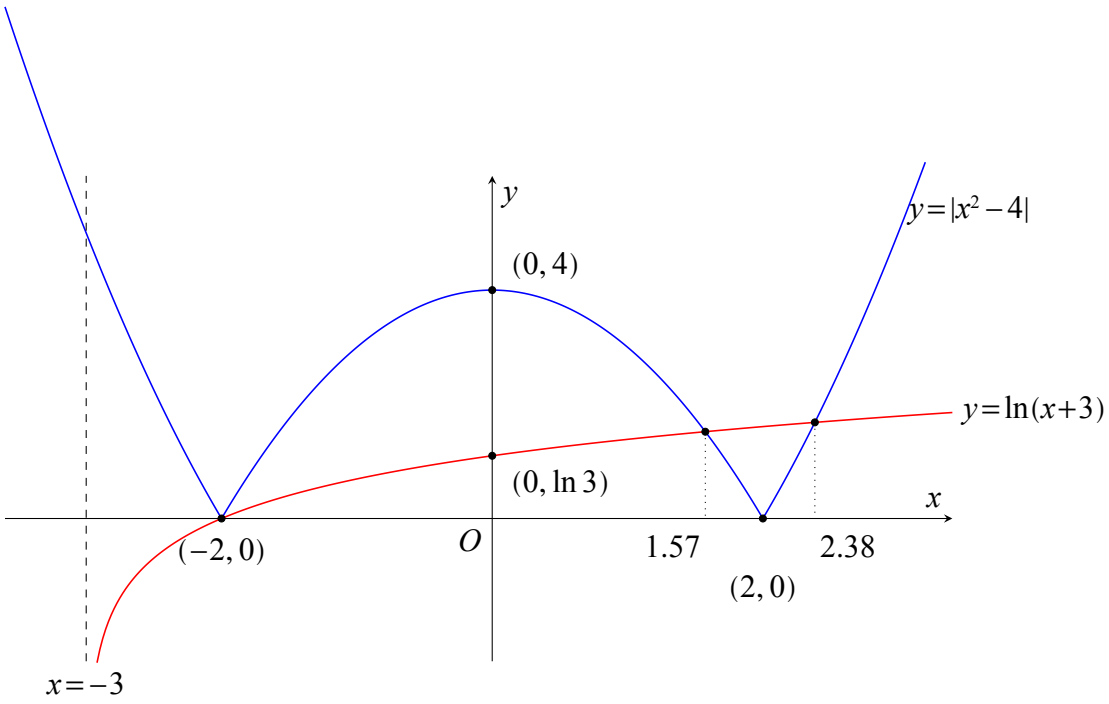
2.1 Solving Inequalities Graphically

Some inequalities cannot be solved algebraically, so we sketch the graphs and read off the solution.

- Rearrange to $f(x) > 0$, or compare $y = f(x)$ with $y = g(x)$.
- Sketch on the GC; mark the intersections and the asymptotes.
- Read off the x-values where the required graph is above or below.

Example 2.1

$$|x^2 - 4| \leqslant \ln(x+3)$$



Using GC, intersections at $x = 1.5746$, $x = 2.3840$

$$\therefore x = -2 \text{ or } 1.57 \leqslant x \leqslant 2.38$$

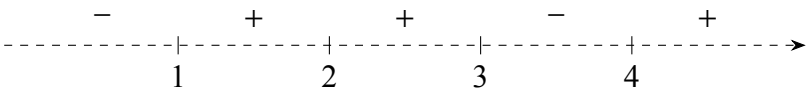
2.2 Sign Test Technique

We solve rational inequalities by finding where the expression changes sign.

- Make the right-hand side 0.
- Do **not** cross multiply unless the expression or bracket is always positive.
- Expressions that are always positive can be divided across, e.g. a quadratic with discriminant <0 and positive leading coefficient: $x^2+2x+5=(x+1)^2+4>0$.

Example 2.2

$$\frac{(x-1)(x-2)^2}{(x-3)(x-4)} \leqslant 0$$



$x=2$: numerator $=0$ (included)

$x=3,\ x=4$: denominator $=0$ (excluded)

$\therefore x \leqslant 1$ or $x=2$ or $3<x<4$

You can sketch the graph to check the answer.

FLAG N1 [MINOR / syllabus] — for Maximus to review: 2.2 Sign Test Technique, worked example: the numerator $(x-1)(x-2)^2$ is a cubic. The 9758 syllabus restricts rational inequalities f/g to linear or quadratic f and g, so this example is beyond syllabus. Consider labelling it as an extension, or using a quadratic numerator.

> **Note**
>
> Crossing a factor with an odd power changes the sign; crossing a factor with an even power keeps the same.

2.3 Common Substitutions

Once an inequality in x is solved, we can replace x with an expression to deduce the solution of a related inequality.

Replace x by	Original inequality		
	$x>2$	$x<-1$	$-1<x<2$
e^x	$x>\ln 2$	no solution	$x<\ln 2$
$\ln x$	$x>e^2$	$0<x<e^{-1}$	$e^{-1}<x<e^2$
$-x$	$x<-2$	$x>1$	$-2<x<1$
$x+1$	$x>1$	$x<-2$	$-2<x<1$
$\lvert x\rvert$	$x<-2$ or $x>2$	no solution	$-2<x<2$
x^2	$x<-\sqrt{2}$ or $x>\sqrt{2}$	no solution	$-\sqrt{2}<x<\sqrt{2}$
$\frac{1}{x}$	$\left(0<x<\frac{1}{2}\right)$	$(-1<x<0)$	$\left(x<-1 \text{ or } x>\frac{1}{2}\right)$
$\cos x,\ 0\leqslant x\leqslant 2\pi$	(no solution)	(no solution)	$(0\leqslant x<\pi \text{ or } \pi<x\leqslant 2\pi)$

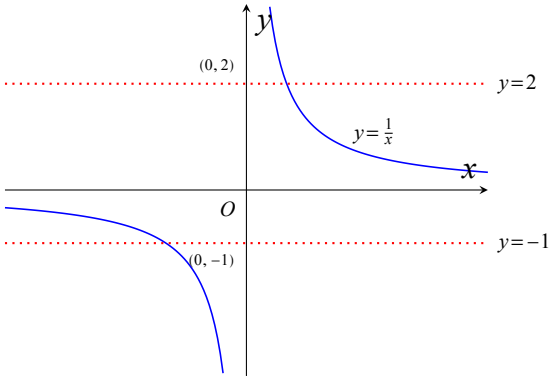
Regardless of whatever substitution you use, using a graph will always work, but some can be solved without a graph.

For $\frac{1}{x}$, $\sin x$ and $\cos x$, draw the graph to solve.

FLAG N2 [MINOR] — for Maximus to review: 2.3 Common Substitutions: the sentence mentions $\sin x$, but the table has no $\sin x$ row (only $\cos x,\ 0\leqslant x\leqslant 2\pi$). Add a $\sin x$ row or remove the mention.

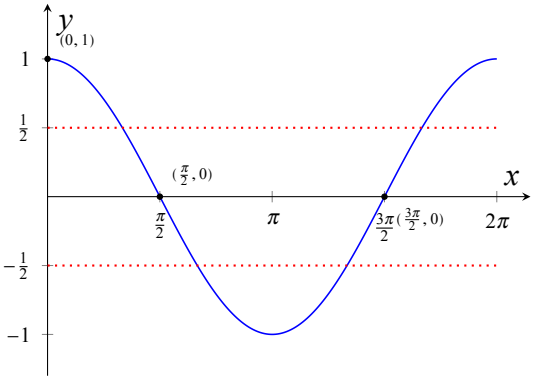
Example 2.3

$$\frac{1}{x}>2, \quad \frac{1}{x}<-1, \quad -1<\frac{1}{x}<2$$



Example 2.4

$$\cos x>\tfrac{1}{2}, \quad \cos x<-\tfrac{1}{2},$$
$$-\tfrac{1}{2}<\cos x<\tfrac{1}{2}, \quad 0\leqslant x\leqslant 2\pi$$

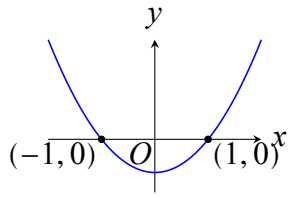


3 Graphing Techniques

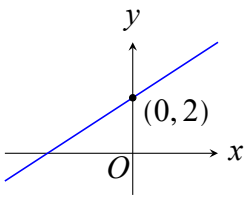
3.1 Key Features of a Curve

Key features of a curve:

(a) **x-intercepts:** let $y=0$.
e.g. $y=x^2-1$: $x^2-1=0 \Rightarrow x=\pm 1$.



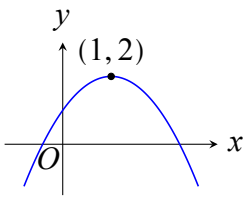
y-intercepts: let $x=0$.
e.g. $y=x+2$: $y=2$.



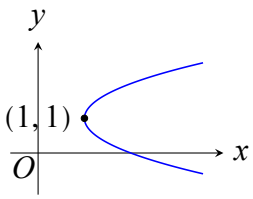
(b) **Stationary points**, when $\frac{dy}{dx}=0$ or $\frac{dy}{dx}$ is undefined. Label upon request.

FLAG N3 [STATEMENT] — for Maximus to review: 3.1(b) Key Features: a stationary point is where $\frac{dy}{dx}=0$ only. A point where $\frac{dy}{dx}$ is undefined (vertical tangent, cusp, endpoint) is not a stationary point; the examples just below list "vertical turning point separately. Reword (e.g. "stationary points: $\frac{dy}{dx}=0$; also note points with a vertical tangent).

Horizontal turning point: $\frac{dy}{dx}=0$ (tangent horizontal).
e.g. $y=-(x-1)^2+2$, maximum point $(1, 2)$.

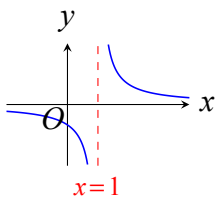
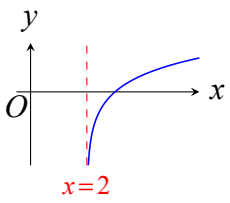
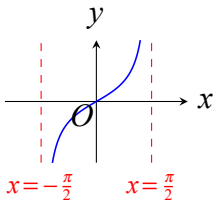


Vertical turning point: $\frac{dy}{dx}$ undefined (tangent vertical).
e.g. $x=(y-1)^2+1$, turning point $(1, 1)$.



(c) Asymptotes

(i) **Vertical asymptotes.** If as $x \to k$, $y \to \pm\infty$, then $x = k$ is a vertical asymptote.

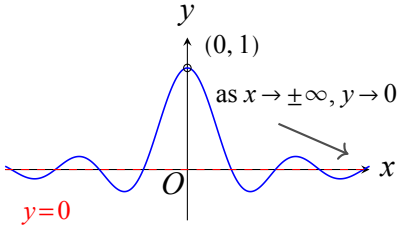
Common Occurrence	**Condition**	**Example and asymptote**	**Graph**
Fractions	denominator $= 0$	$y = \dfrac{1}{x-1}$ $x - 1 = 0 \Rightarrow x = 1$ $\therefore x = 1$ is a vertical asymptote	
Logarithmic graphs	inside ln $= 0$	$y = \ln(x-2)$ $x - 2 = 0 \Rightarrow x = 2$ $\therefore x = 2$ is a vertical asymptote	
Tangent graphs	inside tan $= \dfrac{\pi}{2} + k\pi$, $k \in \mathbb{Z}$	$y = \tan x$ $x = \pm\dfrac{\pi}{2}, \pm\dfrac{3\pi}{2}, \ldots$	

FLAG N4 [MINOR] — for Maximus to review: 3.1(c)(i) Vertical asymptotes table, "Condition column: "denominator $= 0$, "inside ln $= 0$ and "inside tan $= \frac{\pi}{2} + k\pi$ are shorthand. A zero denominator gives a vertical asymptote only if the numerator is non-zero there (otherwise it is a hole), and the log/tan conditions describe the behaviour of the argument as $x \to k$ (e.g. inside $\to 0^+$). Add "(numerator $\neq 0$) and "as $x \to \ldots$ wording.

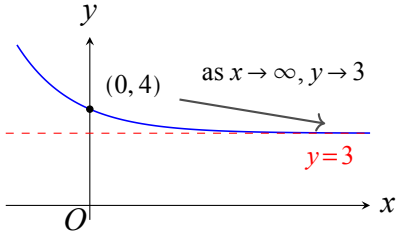
(ii) **Horizontal asymptotes.** As $x \to \pm\infty$, if $y \to k$, then $y = k$ is a horizontal asymptote.

- To find horizontal asymptotes, one should sub in $x \to \pm\infty$ into the equation. This is the only foolproof way to do it.
- A curve **CAN** intersect its horizontal asymptote.

Note: there are more cases of horizontal asymptotes, but these are the common cases.



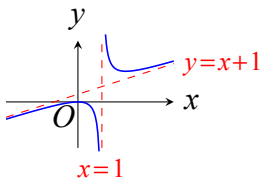
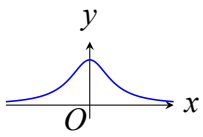
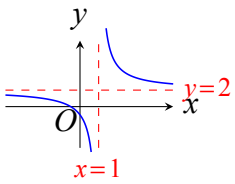
e.g. $y = \dfrac{\sin x}{x}$: curve crosses $y = 0$



e.g. $y = 3 + e^{-x}$

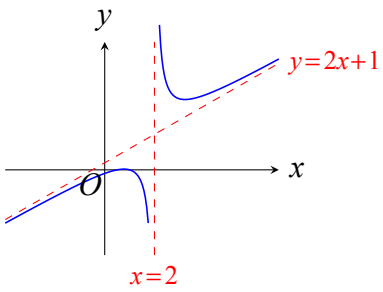
Useful trick to find horizontal asymptotes

For rational functions $\frac{P(x)}{Q(x)}$:

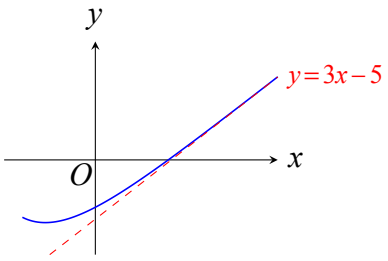
Case	Horizontal asymptote	Example	Graph
numerator power > denominator power	no horizontal asymptote	$y=\frac{x^2}{x-1}$	
numerator power < denominator power	$y=0$	$y=\frac{1}{x^2+1}$	
numerator power = denominator power	ratio of coefficients of highest power	$y=\frac{2x+1}{x-1}$ $y=2$	

(iii) **Oblique asymptotes.** An asymptote of the form $y=mx+c$, $m\neq 0$.

Graphs with oblique asymptotes come in the form $y=mx+c+\mathrm{g}(x)$, where $\mathrm{g}(x)\to 0$ as $x\to\pm\infty$; the oblique asymptote is $y=mx+c$.



e.g. $y=2x+1+\frac{3}{x-2}$



e.g. $y=3x-5+\mathrm{e}^{-x}$

Note

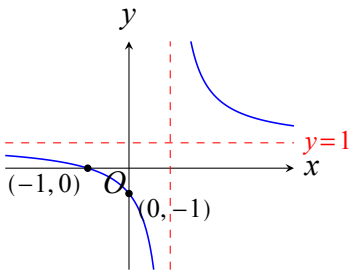
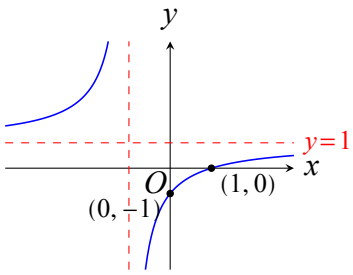
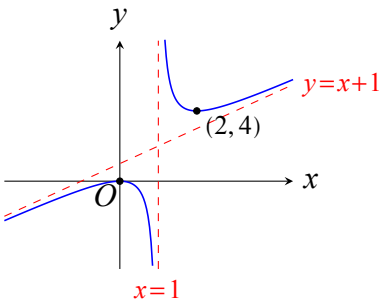
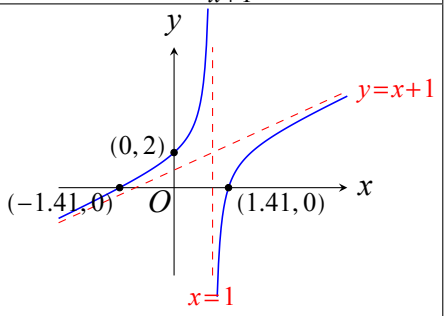
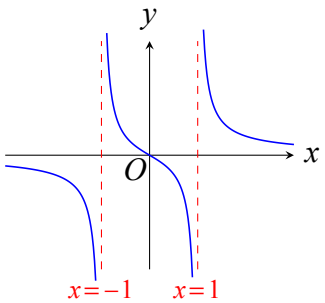
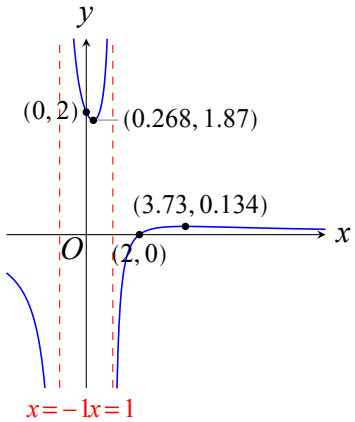
For parametric curves, oblique asymptotes (if they exist) can only be found via the cartesian

FLAG N5 [STATEMENT] — for Maximus to review: 3.1 Note (repeated in 3.5.2): "can only be found via the cartesian equation is too strong. An oblique asymptote can be found directly from the parametric form: $m=\lim y/x$ and $c=\lim(y-mx)$ as $t\to t_0$ (or $t\to\pm\infty$). The cartesian route is just one method.

3.2 Graphs of Rational Functions

Two common shapes for each type.

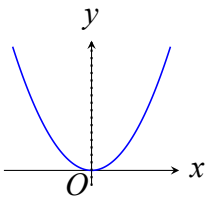
FLAG N6 [MINOR] — for Maximus to review: 3.2 Graphs of Rational Functions: nothing is said about a hole when numerator and denominator share a factor (e.g. $y=\frac{x^2-1}{x-1}$). The shapes and asymptote rules assume no common factor; add a one-line caution.

Type and form	Shape 1	Shape 2
Linear / linear $y=\dfrac{Ax+B}{Cx+D}$ Horizontal asymptote: $y=\dfrac{A}{C}$ Vertical asymptote: $x=-\dfrac{D}{C}$	 $y=\dfrac{x+1}{x-1}$	 $y=\dfrac{x-1}{x+1}$
Quadratic / linear $y=\dfrac{ax^2+bx+c}{dx+e}$ $=Ax+B+\dfrac{C}{dx+e}$ Oblique asymptote: $y=Ax+B$ Vertical asymptote: $x=-\dfrac{e}{d}$	 2 turning points $y=x+1+\dfrac{1}{x-1}\quad\left(\dfrac{Cd}{A}>0\right)$	 No turning points $y=x+1-\dfrac{1}{x-1}\quad\left(\dfrac{Cd}{A}<0\right)$
Linear / quadratic $y=\dfrac{ax+b}{cx^2+dx+e}$ Horizontal asymptote: $y=0$ Vertical asymptotes: solve $cx^2+dx+e=0$ (possibly no roots)	 $y=\dfrac{x}{(x-1)(x+1)}$	 $y=\dfrac{x-2}{(x-1)(x+1)}$

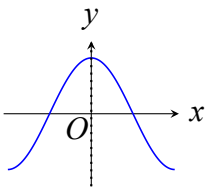
3.3 Symmetry

Even functions: $f(-x)=f(x)$.

The graph is symmetric about the y-axis (line of symmetry $x=0$, dotted).



e.g. $y=x^2$

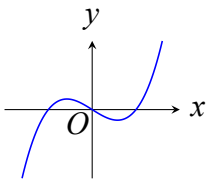


e.g. $y=\cos x$

Odd functions: $f(-x)=-f(x)$.

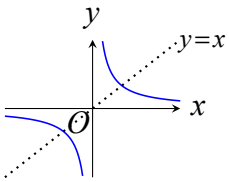
The graph is symmetric about the origin.

e.g. $y=x^3-x$

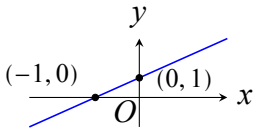
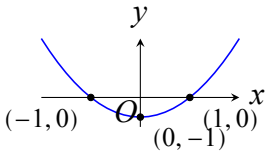
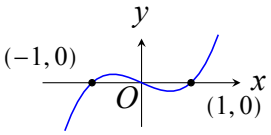
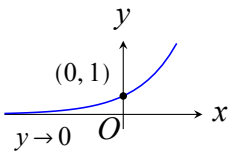
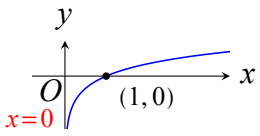
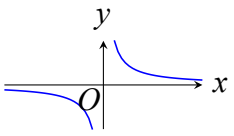
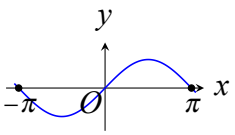
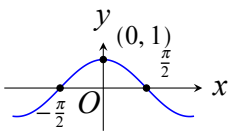
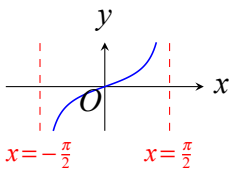
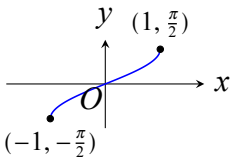
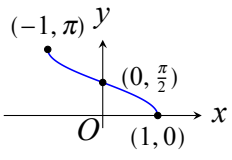
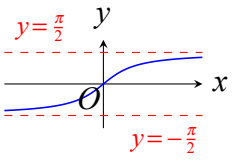


Symmetry about $y=x$: swapping x and y gives the same equation.

e.g. $y=\dfrac{1}{x}$ $(xy=1)$



3.4 Basic Graphs

 Linear $y=x+1$	 Quadratic $y=x^2-1$	 Cubic $y=x^3-x$
 Exponential $y=e^x$	 Logarithm $y=\ln x$	 Reciprocal $y=\frac{1}{x}$
 $y=\sin x$	 $y=\cos x$	 $y=\tan x$
 $y=\sin^{-1}x$	 $y=\cos^{-1}x$	 $y=\tan^{-1}x$

3.5 Parametric Equations

A curve C can be described by giving both coordinates in terms of a parameter t:

$$x=x(t), \quad y=y(t), \quad t\in T$$

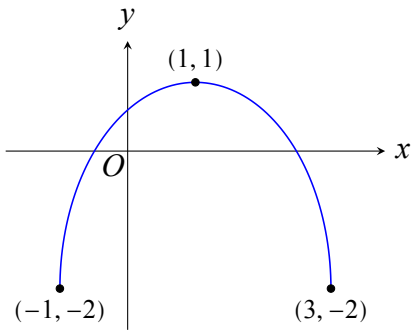
Example 3.1

$x=2\cos t+1, \quad y=3\sin t-2, \quad 0\leqslant t\leqslant \pi$

$t=0$: $(3,-2)$ $\quad t=\frac{\pi}{2}$: $(1,1)$ $\quad t=\pi$: $(-1,-2)$

$x=0$: $\left(0, \frac{3\sqrt{3}}{2}-2\right)$

$y=0$: $\left(1\pm\frac{2\sqrt{5}}{3}, 0\right)$



> **CAUTION: sketching from parametric equations**
>
> **ALWAYS take note of the range of t when sketching graphs.** The restriction on t decides which part of the curve exists: above, $0\leqslant t\leqslant \pi$ gives only the upper half of the ellipse, with endpoints $(3,-2)$ and $(-1,-2)$, not the whole ellipse.

3.5.1 Finding the Cartesian Equation

The cartesian equation of C is an equation without the parameter t. Common techniques:

① **Make t the subject.** e.g. $x=2t-1, y=3t^2+e^t, t\in\mathbb{R}$

$$t=\frac{x+1}{2}$$

$$y=3\left(\frac{x+1}{2}\right)^2+e^{\frac{x+1}{2}}$$

② **Trigo identity formulas.** e.g. $x=2\cos t+1, y=3\sin t-2, 0\leqslant t\leqslant\pi$

$$\cos t=\frac{x-1}{2}, \quad \sin t=\frac{y+2}{3}$$

$$\cos^2 t+\sin^2 t=1$$

$$\frac{(x-1)^2}{4}+\frac{(y+2)^2}{9}=1, \quad y\geqslant -2$$

③ **Others.** e.g. $x=t-\frac{1}{t}, y=t+\frac{1}{t}, t\neq 0$

$$y+x=2t, \quad y-x=\frac{2}{t}$$

$$(y+x)(y-x)=4$$

$$y^2-x^2=4$$

Note: Avoid finding the cartesian eqn if the question did not ask you to.

3.5.2 Asymptotes (Parametric Equations)

- Find (if possible) $t \to t_0$ such that $x \to x_0, y \to \pm\infty$ (vertical asymptote $x = x_0$), or $y \to y_0, x \to \pm\infty$ (horizontal asymptote $y = y_0$).
- Oblique asymptotes (if they exist) can only be found via the cartesian equation.

Example 3.2

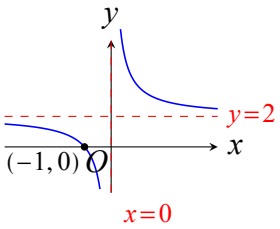
$x = \dfrac{1}{t-1},\ y = 2t,\ t \neq 1$

As $t \to \pm\infty,\ x \to 0,\ y \to \pm\infty$

$\therefore x = 0$ is a vertical asymptote

As $t \to 1,\ x \to \pm\infty,\ y \to 2$

$\therefore y = 2$ is a horizontal asymptote



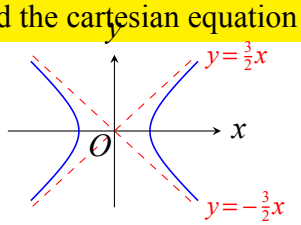
Oblique asymptotes.

Cannot be found from the parametric form alone; find the cartesian equation first.

e.g. $x = 2\sec\theta,\ y = 3\tan\theta$

$$\sec^2\theta - \tan^2\theta = 1$$

$$\frac{x^2}{4} - \frac{y^2}{9} = 1$$



Oblique asymptotes: $y = \pm\dfrac{3}{2}x$

FLAG N7 [STATEMENT] — for Maximus to review: 3.5.2 Asymptotes (Parametric Equations): same issue as the 3.1 Note — "Cannot be found from the parametric form alone and "can only be found via the cartesian equation are too strong; $m = \lim y/x,\ c = \lim(y - mx)$ can be taken straight from $x(t), y(t)$.

3.5.3 Intersections

Intersection between a parametric curve and another curve: sometimes both equations are needed.

e.g. $x = 2t + \ln t,\ y = t^2 - 1,\ t > 0$ and the circle $x^2 + y^2 = 25$:

$$(2t + \ln t)^2 + (t^2 - 1)^2 = 25$$

Using GC, $t = 0.00735$ or $t = 1.86$ (both > 0)

$\therefore$ intersections: $(-4.90,\ -1.00)$ and $(4.35,\ 2.47)$

This equation in t cannot be solved algebraically, so the GC is needed.

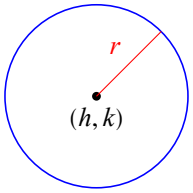
3.6 Conics

Circle

$$(x-h)^2+(y-k)^2=r^2$$

Centre: (h,k)

Radius: r

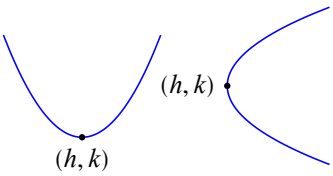


Parabola

$$y-k=a(x-h)^2 \qquad x-h=b(y-k)^2$$

$y-k=a(x-h)^2$:
turning point (h,k), $\dfrac{dy}{dx}=0$

$x-h=b(y-k)^2$:
turning point (h,k), $\dfrac{dy}{dx}$ undefined



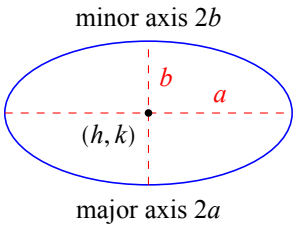
Ellipse

$$\frac{(x-h)^2}{a^2}+\frac{(y-k)^2}{b^2}=1$$

Centre: (h,k)

"Horizontal radius" $=a$, "vertical radius" $=b$

For $a>b$: major axis horizontal (length $2a$, semi-major a); minor axis vertical (length $2b$, semi-minor b).



> ⋆ RHS ⇒ numerator coefficients of x^2, y^2 must all be 1.

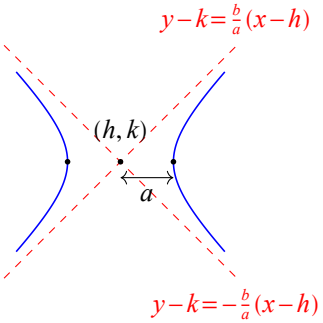
Hyperbola: Type I

$$\frac{(x-h)^2}{a^2}-\frac{(y-k)^2}{b^2}=1$$

Centre: (h,k)

Distance from centre to vertices $=a$

Asymptotes (dashed): $y-k=\pm\frac{b}{a}(x-h)$



> ⋆ RHS ⇒ numerator coefficients of x^2, y^2 must all be 1.
> ⋆ Equation of asymptotes: $y-k=\pm\frac{b}{a}(x-h)$

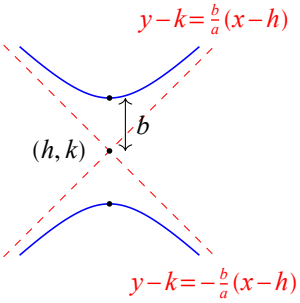
Hyperbola: Type II

$$\frac{(y-k)^2}{b^2}-\frac{(x-h)^2}{a^2}=1$$

Centre: (h,k)

Distance from centre to vertices $=b$

Asymptotes (dashed): $y-k=\pm\frac{b}{a}(x-h)$



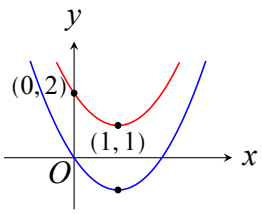
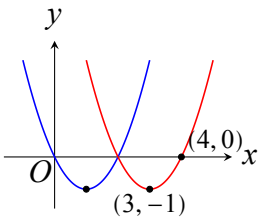
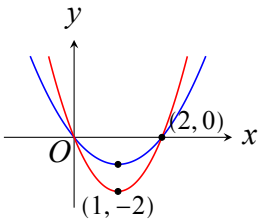
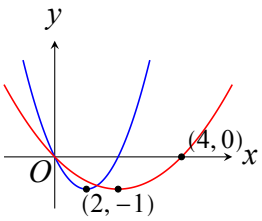
> ⋆ RHS ⇒ numerator coefficients of x^2, y^2 must all be 1.
> ⋆ Equation of asymptotes: $y-k=\pm\frac{b}{a}(x-h)$

4 Graphing Transformations

Every transformation comes from **replacing** x **or** y in the equation $y=\mathrm{f}(x)$.

Throughout, blue is $y=\mathrm{f}(x)=x(x-2)$, with intercepts $(0,0)$, $(2,0)$ and minimum point $(1,-1)$; red is the transformed graph.

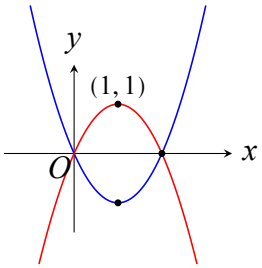
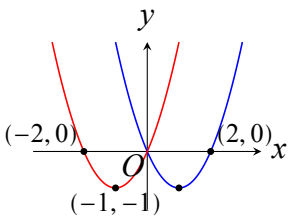
4.1 Translations and Scalings

Transformation	Replacement	Result	Graph
Translation of a units in the positive y direction	replace y with $y-a$: $y-a=\mathrm{f}(x)$	$y=\mathrm{f}(x)+a$ (graph: $a=2$)	
Translation of a units in the positive x direction	replace x with $x-a$	$y=\mathrm{f}(x-a)$ (graph: $a=2$)	
Scaling by factor a parallel to the y-axis	replace y with $\frac{y}{a}$: $\frac{y}{a}=\mathrm{f}(x)$	$y=a\mathrm{f}(x)$ (graph: $a=2$)	
Scaling by factor a parallel to the x-axis	replace x with $\frac{x}{a}$	$y=\mathrm{f}\left(\frac{x}{a}\right)$ (graph: $a=2$)	

CAUTION

Every replacement acts the "opposite" way to what it looks like: $x-2$ moves the graph in the **positive** x direction; $y=\mathrm{f}(2x)$ is a scaling by factor $\frac{1}{2}$ (not 2) parallel to the x-axis.
Replacing x does **not** mean replacing the input of f, and replacing y does not mean replacing the output $\mathrm{f}(x)$.
If you are ever confused, think of $\sin x$ vs $\sin 2x$.

4.2 Reflections

Transformation	Replacement	Result	Graph
Reflection about the x-axis	replace y with $-y$: $-y=\mathrm{f}(x)$	$y=-\mathrm{f}(x)$	
Reflection about the y-axis	replace x with $-x$	$y=\mathrm{f}(-x)$	

4.3 Combining Transformations

Functions of the form $y=a\mathrm{f}(bx+c)+d,\ \ a,b,c,d\in\mathbb{R}$: replace x and y **one step at a time**.

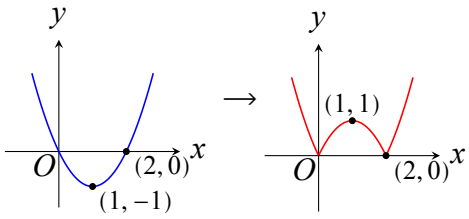
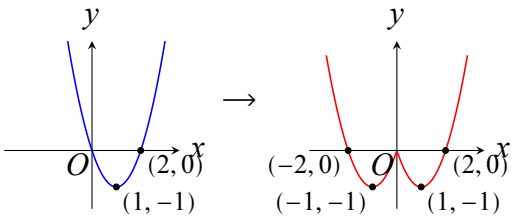
Example 4.1

$y=\mathrm{f}(x)\ \longrightarrow\ y=2\mathrm{f}(2x-1)+1$

Replacement	Transformation	Function
replace x with $x-1$	translation of 1 unit in the positive x direction	$y=\mathrm{f}(x-1)$
replace x with $2x$	scaling by factor $\frac{1}{2}$ parallel to the x-axis	$y=\mathrm{f}(2x-1)$
replace y with $\frac{y}{2}$	scaling by factor 2 parallel to the y-axis	$y=2\mathrm{f}(2x-1)$
replace y with $y-1$	translation of 1 unit in the positive y direction	$y=2\mathrm{f}(2x-1)+1$

4.4 Modulus Graphs

Every modulus graph can be partitioned into a "+" part and a "−" part: $|x|=x$ for $x\geqslant 0$, $|x|=-x$ for $x<0$.

Graph	How to sketch	Graph
$y=\|f(x)\|$	Keep the part on or above the x-axis; reflect the part below the x-axis about the x-axis. Sharp points at the x-intercepts.	
$y=f(\|x\|)$	Remove the part on the left of the y-axis; reflect the part on the right of the y-axis about the y-axis.	

FLAG N8 [MINOR] — for Maximus to review: 4.4 Modulus Graphs, row $y=|f(x)|$: "Sharp points at the x-intercepts is not always true. Where the curve only touches the axis (repeated root, e.g. $|x^2|$) or crosses with zero gradient (e.g. $|x^3|$) there is no sharp point. Also $y=f(|x|)$ usually has a sharp point on the y-axis, which the second row does not mention.

Example 4.2

$y=f(x) \longrightarrow y=|f(-2x+1)|$

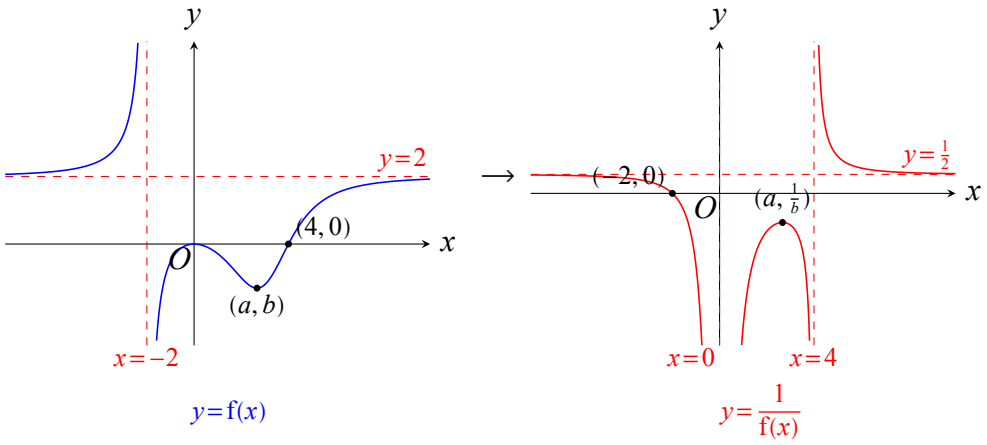
Replacement	Transformation	Function
replace x with $x+1$	translation of 1 unit in the negative x direction	$y=f(x+1)$
replace x with $-x$	reflection about the y-axis	$y=f(-x+1)$
replace x with $2x$	scaling by factor $\frac{1}{2}$ parallel to the x-axis	$y=f(-2x+1)$
$\|f\|$	reflect parts below the x-axis about the x-axis	$y=\|f(-2x+1)\|$

4.5 Reciprocal Graphs $y=\frac{1}{\mathrm{f}(x)}$

Analyse the new curve at the critical points of $y=\mathrm{f}(x)$.

x-intercept $(a,0)$	$\longleftrightarrow$	vertical asymptote $x=a$
vertical asymptote $x=a$	$\longleftrightarrow$	$y\to 0$: point $(a,0)$ (excluded)
maximum point (a,b), $b\neq 0$	$\longleftrightarrow$	minimum point $\left(a,\frac{1}{b}\right)$
minimum point (a,b), $b\neq 0$	$\longleftrightarrow$	maximum point $\left(a,\frac{1}{b}\right)$
horizontal asymptote $y=k$, $k\neq 0$	$\longleftrightarrow$	horizontal asymptote $y=\frac{1}{k}$
$y\to\pm\infty$ (incl. oblique asymptote)	$\longleftrightarrow$	horizontal asymptote $y=0$
$y=\pm 1$	$\longleftrightarrow$	same point $(\frac{1}{\pm 1}=\pm 1)$
$\mathrm{f}(x)>0$ / $\mathrm{f}(x)<0$	$\longleftrightarrow$	$\frac{1}{\mathrm{f}(x)}>0$ / $\frac{1}{\mathrm{f}(x)}<0$ (same sign)

FLAG N9 [MINOR] — for Maximus to review: 4.5 Reciprocal Graphs, table: the row "vertical asymptote $x=a \leftrightarrow y\to 0$: point $(a,0)$ (excluded) is loosely worded — $\frac{1}{\mathrm{f}}\to 0$ as $x\to a$, so the curve approaches the x-axis near $x=a$ (it does not pass through a point $(a,0)$). There is also no row for "horizontal asymptote $y=0 \leftrightarrow y\to\pm\infty$.



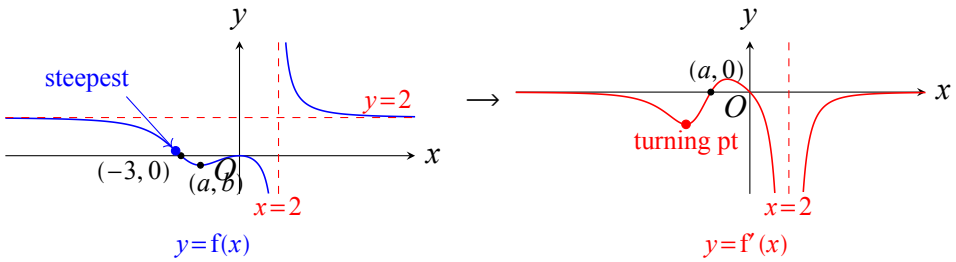
Note: sketching $y=\frac{1}{\mathrm{f}(x)}$

- Same sign: $\mathrm{f}>0\Rightarrow\frac{1}{\mathrm{f}}>0$, $\mathrm{f}<0\Rightarrow\frac{1}{\mathrm{f}}<0$.
- $|\mathrm{f}|$ large $\Rightarrow\left|\frac{1}{\mathrm{f}}\right|$ small; $|\mathrm{f}|$ small $\Rightarrow\left|\frac{1}{\mathrm{f}}\right|$ large.
- $\mathrm{f}=0\Rightarrow\frac{1}{\mathrm{f}}$ undefined (vertical asymptote).
- $\mathrm{f}=\pm 1\Rightarrow\frac{1}{\mathrm{f}}=\pm 1$ (graphs meet).

4.6 Derivative Graphs $y=\mathrm{f}'(x)$

$\mathrm{f}'(x)=\dfrac{\mathrm{d}y}{\mathrm{d}x}$ is the gradient of $y=\mathrm{f}(x)$: read off the gradient on each interval.

$y=\mathrm{f}(x)$	$y=\mathrm{f}'(x)$
stationary point at $x=a$	x-intercept $(a,0)$
increasing / decreasing	above / below the x-axis
point of inflexion (max. or min. gradient)	turning point
vertical asymptote $x=a$	vertical asymptote $x=a$
horizontal asymptote $y=k$	horizontal asymptote $y=0$
oblique asymptote $y=mx+c$	horizontal asymptote $y=m$



i The turning point on the derivative graph.

CAUTION

i Mark x-coordinates only where they come from $y=\mathrm{f}(x)$: the y-values of $y=\mathrm{f}'(x)$ are generally **not** known from the graph of $y=\mathrm{f}(x)$.

FLAG N10 [MINOR] — for Maximus to review: 4.6 Derivative Graphs, info box: "steepest point only describes a maximum of $y=\mathrm{f}'(x)$ (greatest gradient). A minimum of $y=\mathrm{f}'(x)$ is the least gradient (e.g. the steepest downhill point if negative). Better: "a turning point of $y=\mathrm{f}'(x)$ corresponds to a point of inflexion (greatest or least gradient) on $y=\mathrm{f}(x)$.

4.7 Reverse Transformations

If $y=\mathrm{f}(x)$ undergoes a sequence of transformations to give a known $y=\mathrm{g}(x)$, undo the transformations **in reverse order**, each with its opposite replacement.

Example 4.3

$y=\mathrm{f}(x)$ undergoes, in order, the following to give $y=6\mathrm{e}^{-x+3}$. Find $\mathrm{f}(x)$.

A: translation of 2 units in the negative x direction

B: reflection about the y-axis

C: scaling by factor 3 parallel to the y-axis

Step	Opposite transformation	Replacement	Function
C^{-1}	scale by factor $\frac{1}{3}$ parallel to the y-axis	replace y with $3y$	$y=2e^{-x+3}$
B^{-1}	reflect about the y-axis	replace x with $-x$	$y=2e^{x+3}$
A^{-1}	translate 2 units in the positive x direction	replace x with $x-2$	$y=2e^{x+1}$

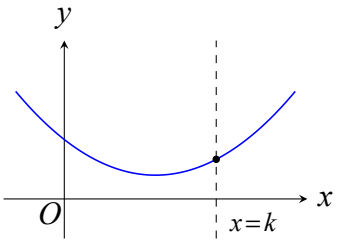
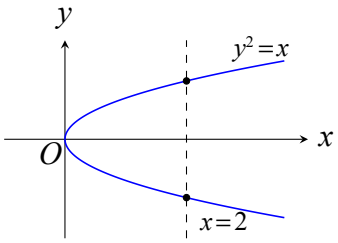
$\therefore f(x)=2e^{x+1}$

5 Functions

5.1 Functions, Domain and Range

$$\mathrm{f}: x \mapsto \underbrace{x^2+3x+2}_{\text{rule f}(x)}, \quad x \in \mathbb{R}, \quad \underbrace{x \geqslant 0}_{\text{domain } D_\mathrm{f}}$$

A function is a relation that gives **exactly one** value of y for each $x \in D_\mathrm{f}$.

It's a function	Not a function
 every vertical line $x=k$, $k \in D_\mathrm{f}$, cuts the graph **exactly once**	 one vertical line ($x=2$) cuts the graph **more than once**

Inequality	Set notation	Interval
$a<x<b$	$\{x \in \mathbb{R} : a<x<b\}$	(a, b)
$a \leqslant x \leqslant b$	$\{x \in \mathbb{R} : a \leqslant x \leqslant b\}$	$[a, b]$
$a \leqslant x<b$	$\{x \in \mathbb{R} : a \leqslant x<b\}$	$[a, b)$
$x>a$	$\{x \in \mathbb{R} : x>a\}$	(a, ∞)
$x \in \mathbb{R},\ x \neq 0$	$\mathbb{R} \setminus \{0\}$	$(-\infty, 0) \cup (0, \infty)$

CAUTION

To find the range of a function, **ALWAYS** sketch its graph $y=\mathrm{f}(x)$, $x \in D_\mathrm{f}$, and read the range off the graph. **Never** just sub in the endpoints of the domain.

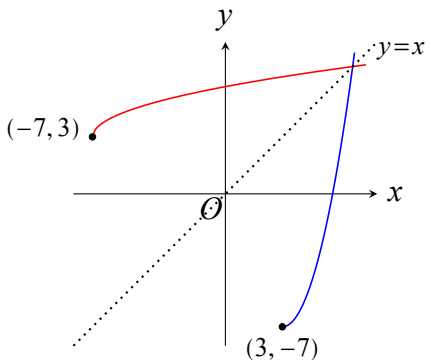
5.2 Inverse Functions

	Working
Exists	**Exists:** f is one-to-one: every line $y=k,\ k\in\mathbb{R}$, cuts $y=\mathrm{f}(x)$ **at most once** $\Rightarrow \mathrm{f}^{-1}$ exists. $\Rightarrow$ f is one-to-one, thus f^{-1} exists. **Does not exist:** e.g. $\mathrm{f}(x)=(x-3)^2-7,\ x\in\mathbb{R}$: the line $y=0$ cuts $y=\mathrm{f}(x)$ **twice** (at $x=3\pm\sqrt{7}$) $\Rightarrow \mathrm{f}^{-1}$ does not exist. Alternatively: $\exists\, x_1\neq x_2$ such that $\mathrm{f}(x_1)=\mathrm{f}(x_2)$ $\therefore$ f is not one-one $\Rightarrow \mathrm{f}^{-1}$ does not exist.
Rule	make x the subject of $y=\mathrm{f}(x)$
Domain	$D_{\mathrm{f}^{-1}}=R_{\mathrm{f}}$
Range	$R_{\mathrm{f}^{-1}}=D_{\mathrm{f}}$
Graph	reflection of $y=\mathrm{f}(x)$ about $y=x$: all x- and y-coordinates are swapped, $(a,b)\to(b,a)$

Example 5.1

$\mathrm{f}: x\mapsto (x-3)^2-7,\ x\geqslant 3$

$$y=(x-3)^2-7$$
$$x-3=\sqrt{y+7}\quad (x\geqslant 3)$$
$$x=3+\sqrt{y+7}$$
$$R_{\mathrm{f}}=[-7,\infty)$$
$$\therefore \mathrm{f}^{-1}: x\mapsto 3+\sqrt{x+7},\ x\in\mathbb{R},\ x\geqslant -7$$



CAUTION

- When drawing $y=\mathrm{f}(x)$ and $y=\mathrm{f}^{-1}(x)$ together, use the **same scale** on both axes.
- $\mathrm{ff}^{-1}(x)=\mathrm{f}^{-1}\mathrm{f}(x)=x$, but $D_{\mathrm{ff}^{-1}}=D_{\mathrm{f}^{-1}}=R_{\mathrm{f}}$ while $D_{\mathrm{f}^{-1}\mathrm{f}}=D_{\mathrm{f}}$.
- Solving $\mathrm{f}(x)=\mathrm{f}^{-1}(x)$: it is **not always** $\mathrm{f}(x)=x$ (only when the curves meet on $y=x$).

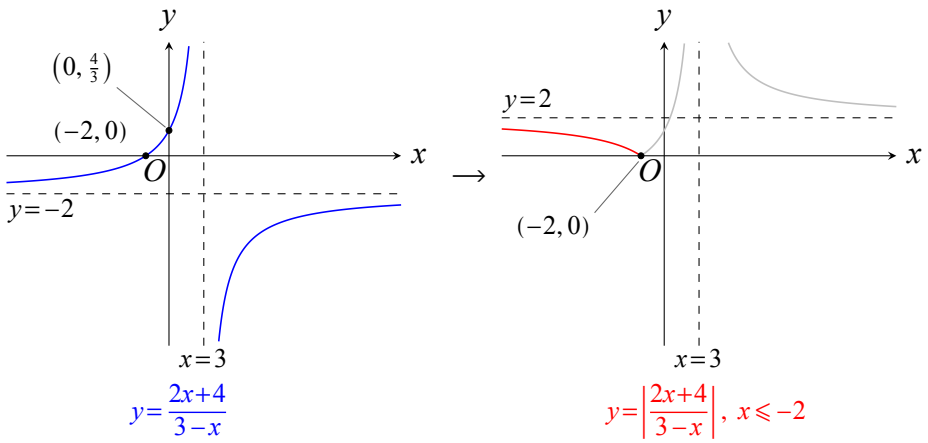
5.2.1 Removing the Modulus

On a restricted domain, the sign inside the modulus is fixed, so the modulus can be removed: $|x-a| = x-a$ for $x \geqslant a$, $|x-a| = -(x-a)$ for $x < a$.

Example 5.2

$\mathrm{f}: x \to \left|\dfrac{2x+4}{3-x}\right|$, $x \in \mathbb{R}$, $x \leqslant -2$ (2023 A-Level P1 Q7, with the domain restricted to $x \leqslant a$, $a = -2$)

$$y = \mathrm{f}(x) = \left|\frac{2x+4}{3-x}\right|$$
$$= -\frac{2x+4}{3-x} \qquad (x \leqslant -2)$$



Note

Whenever you are asked to find f^{-1} and there is a modulus, you are usually able to remove it in the first step by determining whether the region of the graph is positive or negative for the domain of f.

5.3 Composite Functions

	Working
Exists	**Exists:** $R_g \subseteq D_f$ **Does not exist:** $x_0 \in R_g, x_0 \notin D_f \therefore R_g \not\subseteq D_f$
Rule	sub $g(x)$ into $f(x)$: $f[g(x)]$
Domain	$D_{fg} = D_g$: rightmost function's domain (e.g. $D_{fgh} = D_h$)
Range	$D_g \xrightarrow{g} R_g \xrightarrow{f} R_{fg}$: sketch $y = f(x)$ on the domain R_g Alternatively, sketch the graph of $fg(x)$ for D_{fg} (avoid, as fg might usually be quite hard to deal with).

5.3.1 Composite with a Piecewise Function

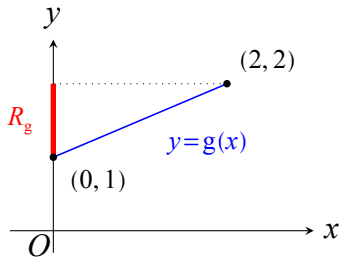
Split according to which piece of f the value $g(x)$ lands in.

Example 5.3

$$f(x) = \begin{cases} x^2, & 0 \leqslant x < 1 \\ 2-x, & 1 \leqslant x \leqslant 2 \end{cases} \qquad g(x) = \frac{x}{2} + 1, \ 0 \leqslant x \leqslant 2$$

$$R_g = [1, 2]$$

$$\therefore fg(x) = f(g(x)) = 2 - \left(\frac{x}{2} + 1\right) = 1 - \frac{x}{2}, \quad 0 \leqslant x \leqslant 2$$

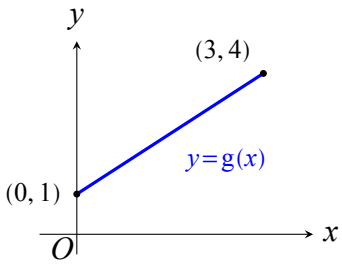


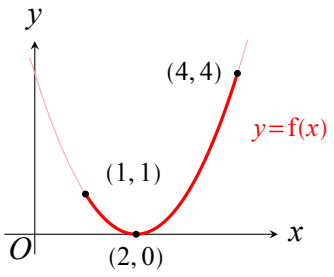
Example 5.4

$g(x) = x + 1, \ 0 \leqslant x \leqslant 3; \quad f(x) = (x-2)^2, \ x \in \mathbb{R}$. Assume fg exists.

$$\begin{array}{ccccc} D_g & \xrightarrow{g} & R_g & \xrightarrow{f} & R_{fg} \\ [0, 3] & & [1, 4] & & [0, 4] \end{array}$$

$$\therefore R_{fg} = [0, 4]$$





5.4 Functional Powers and Self-Inverse Functions

$\mathrm{f}^2(x)=\mathrm{f}[\mathrm{f}(x)]$ exists if $R_\mathrm{f}\subseteq D_\mathrm{f}$; similarly f^3, f^4, … Note $\mathrm{f}^2(x)\neq[\mathrm{f}(x)]^2$.

Self-inverse: $\mathrm{f}=\mathrm{f}^{-1}\iff \mathrm{f}^2(x)=x$ for all $x\in D_\mathrm{f}$ (graph symmetric about $y=x$).

FLAG N11 [MINOR] — for Maximus to review: 5.4 Self-inverse: make the domain condition explicit. $\mathrm{f}=\mathrm{f}^{-1}$ needs $D_\mathrm{f}=R_\mathrm{f}$ (same domain and range), which is what lets f^2 exist on all of D_f. The statement only mentions "for all $x\in D_\mathrm{f}$, so a reader may forget to check $R_\mathrm{f}\subseteq D_\mathrm{f}$ (and $D_\mathrm{f}=R_\mathrm{f}$) before concluding.

Periodic composition pattern. If $\mathrm{g}^2(x)=x$:

$$\mathrm{g}^n(x)=\begin{cases}\mathrm{g}(x), & n \text{ odd}\\ x, & n \text{ even}\end{cases}$$

$$\mathrm{g}^{2027}(x)=\mathrm{g}(x).$$

Similarly, if $\mathrm{f}^3(x)=x$, the pattern repeats every 3: $\mathrm{f}^n=\mathrm{f}^{n \bmod 3}$, and $\mathrm{f}^{-1}=\mathrm{f}^2$.

5.5 Piecewise and Periodic Functions

$$\mathrm{f}:x\mapsto\begin{cases}\mathrm{f}_1(x), & x\in D_1\\ \mathrm{f}_2(x), & x\in D_2\end{cases},\qquad \mathrm{f}(x)=\mathrm{f}(x+p)$$

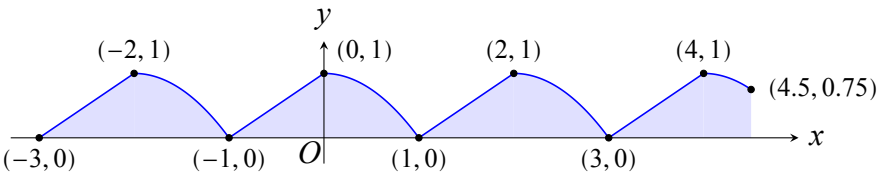
$\mathrm{f}(x)=\mathrm{f}(x+p)$: the graph repeats every p units (period p).

→ Always sketch the **original copy** first, then copy & paste.

→ Never sub any x value into f that is outside the domain of the original copy: shift by multiples of p first.

Example 5.5

$$\mathrm{f}(x)=\begin{cases}1-x^2, & 0\leqslant x<1\\ x-1, & 1\leqslant x<2\end{cases},\ \mathrm{f}(x)=\mathrm{f}(x+2),\ -3\leqslant x\leqslant 4.5$$



$$\mathrm{f}(4.5)=\mathrm{f}(4.5-4)=\mathrm{f}(0.5)=1-0.5^2=0.75$$

Area under a periodic function: area over n full periods $=n\times$ area over one period.

$$\begin{aligned}\int_{-3}^{4.5}\mathrm{f}(x)\,\mathrm{d}x&=\int_{-3}^{3}\mathrm{f}(x)\,\mathrm{d}x+\int_{3}^{4}\mathrm{f}(x)\,\mathrm{d}x+\int_{4}^{4.5}\mathrm{f}(x)\,\mathrm{d}x\\&=3\int_{0}^{2}\mathrm{f}(x)\,\mathrm{d}x+\int_{1}^{2}(x-1)\,\mathrm{d}x+\int_{0}^{0.5}(1-x^2)\,\mathrm{d}x\\&=\frac{107}{24}\approx 4.46\end{aligned}$$

5.5.1 Inverse of a Piecewise Function

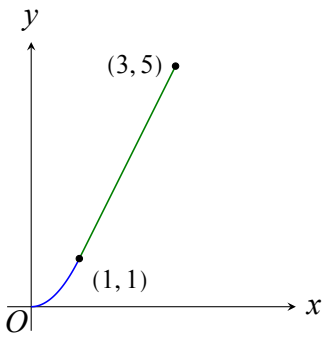
Check one-to-one using the ranges of the pieces (they must not overlap and each piece is one-to-one); then invert each piece, with domain = range of that piece.

Example 5.6

$$\mathrm{h}(x)=\begin{cases} x^2, & 0\leqslant x\leqslant 1 \\ 2x-1, & 1<x\leqslant 3 \end{cases}$$

ranges: $[0,1]$ and $(1,5]$ (no overlap)

$$\mathrm{h}^{-1}(x)=\begin{cases} \sqrt{x}, & 0\leqslant x\leqslant 1 \\ \dfrac{x+1}{2}, & 1<x\leqslant 5 \end{cases}$$



6 Sequences and Series

A **sequence** is a set of numbers arranged in a definite order, A **series** is the sum of the terms of a sequence:

$$u_1,\ u_2,\ u_3,\ \ldots,\quad u_n\text{: the } n\text{th term}$$

$$S_n = u_1 + u_2 + \cdots + u_n = \sum_{r=1}^{n} u_r, \qquad S_\infty = u_1 + u_2 + u_3 + \cdots = \sum_{r=1}^{\infty} u_r \quad \text{(infinite series)}$$

FLAG N12 [OMISSION] — for Maximus to review: Ch 6 Sequences and Series: there is no section on the method of differences (summation of series by the method of differences is listed in the 9758 syllabus, topic 2.2), although Example 6.5.1 (Substitution) relies on a result obtained that way. Consider adding it.

6.1 Recurrence Relations

A recurrence relation defines u_n in terms of one or more preceding terms, together with the initial condition(s). **First-order recurrence relation**: requires only one initial condition; defined using only the immediately preceding term, i.e. u_{n-1} only.

Second-order recurrence relation: requires two initial conditions; defined using the two preceding terms, i.e.

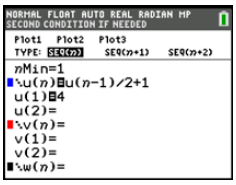
FLAG N13 [MINOR] — for Maximus to review: 6.1 Recurrence Relations: (1) the 9758 syllabus does not list recurrence relations explicitly, and second-order ones are not in the past-paper questions collected in this document — check that second-order is intended. (2) Notation: this paragraph uses u_{n-1}, u_{n-2}, whereas Example 6.3 below uses u_{n+2}, u_{n+1}, u_n; pick one convention.

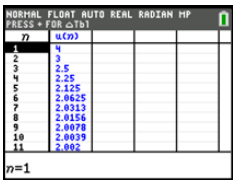
Example 6.1

$$u_{r+1} = \frac{u_r}{2} + 1, \quad u_1 = 4$$

$$u_2 = 3, \quad u_3 = \frac{5}{2}, \quad u_4 = \frac{9}{4}$$

Rule: $u_{r+1} = \frac{u_r}{2} + 1$; initial condition: $u_1 = 4$.





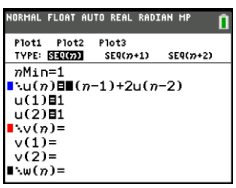
Example 6.2

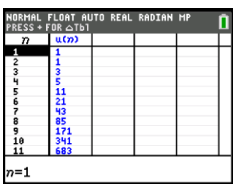
$$u_{r+1} = 2u_r + 1, \quad u_1 = 1$$

Example 6.3

$$u_{n+2} = u_{n+1} + 2u_n, \quad u_1 = 1, \quad u_2 = 1$$

$$1,\ 1,\ 3,\ 5,\ 11,\ 21,\ 43,\ \ldots$$





Example 6.4

(GP)

$$u_{n+1}=ru_n,\quad u_1=a$$

$$a,\ ar,\ ar^2,\ \ldots$$

Example 6.5

(AP)

$$u_{n+1}=u_n+d,\quad u_1=a$$

$$a,\ a+d,\ a+2d,\ \ldots$$

(However, this is not unique.)

$$u_{n+2}-2u_{n+1}+u_n=0,\quad u_1=a,\quad u_2=a+d$$

6.1.1 Limit of a Sequence

$$\text{As } r\to\infty,\ u_r=\frac{1}{r}\to 0 \quad \text{or} \quad \lim_{r\to\infty}(u_r)=\lim_{r\to\infty}\left(\frac{1}{r}\right)=0$$

$$\text{As } r\to\infty,\ u_r\to L \quad \text{or} \quad \lim_{r\to\infty}(u_r)=L \quad \text{(convergent)}$$

$u_r\to L \Rightarrow u_{r+1}\to L,\ u_{r+2}\to L,\ \ldots$

(For recurrence relations: if the sequence is given to be convergent, change all u_n, u_{n+1}, u_{n+2} to L and solve for L.)

Rules ($\lim u_r$, $\lim v_r$ real):

$$\lim_{r\to\infty}(a)=a$$

$$\lim_{r\to\infty}(u_r\pm v_r)=\lim_{r\to\infty}(u_r)\pm\lim_{r\to\infty}(v_r)$$

$$\lim_{r\to\infty}(u_rv_r)=\lim_{r\to\infty}(u_r)\times\lim_{r\to\infty}(v_r)$$

$$\lim_{r\to\infty}\left(\frac{u_r}{v_r}\right)=\frac{\lim_{r\to\infty}(u_r)}{\lim_{r\to\infty}(v_r)},\quad \lim_{r\to\infty}(v_r)\neq 0$$

6.2 Sigma Notation

$$\sum_{r=m}^{n}\mathrm{f}(r)=\mathrm{f}(m)+\mathrm{f}(m+1)+\cdots+\mathrm{f}(n)$$

$$S_n=\sum_{r=1}^{n}u_r=u_1+u_2+\cdots+u_n$$ (By default, S_n starts from $r=1$ unless otherwise stated.)

Properties:

$$\text{no. of terms in } \sum_{r=m}^{n}u_r:\ n-m+1$$

$$\sum_{r=m}^{n}a=(n-m+1)a$$

$$\sum_{r=m}^{n}au_r=a\sum_{r=m}^{n}u_r$$

$$\sum_{r=m}^{n}(u_r\pm v_r)=\sum_{r=m}^{n}u_r\pm\sum_{r=m}^{n}v_r$$

$$\sum_{r=m}^{n}u_r=\sum_{r=1}^{n}u_r-\sum_{r=1}^{m-1}u_r$$

(To remember the last three, note that the properties are similar to what you can do in integration.)

6.3 Standard Formulas

$$\sum_{r=1}^{n} r = \frac{n(n+1)}{2}$$

AP with first term 1, common difference 1:

$$\sum_{r=1}^{n} r = 1+2+\cdots+n$$

$$S_n = \frac{n}{2}[2(1)+(n-1)(1)]$$

$$= \frac{n(n+1)}{2}$$

$$\sum_{r=1}^{n} a^r = \frac{a(a^n-1)}{a-1}, \quad a \neq 1$$

GP with first term a, common ratio a:

$$\sum_{r=1}^{n} a^r = a+a^2+\cdots+a^n$$

$$= \frac{a(a^n-1)}{a-1}$$

$$\sum_{r=1}^{n} r^2 = \frac{n(n+1)(2n+1)}{6}$$

$$\sum_{r=1}^{n} r^3 = \left[\frac{n(n+1)}{2}\right]^2 = \frac{n^2(n+1)^2}{4}$$

Note

$\sum r^2$ and $\sum r^3$ are given in the exam if needed. All four results hold only if r starts from 1.

6.4 Augmented Forms

Example 6.6

$$\sum_{r=1}^{n} \frac{4}{5^{2r+1}} = 4\sum_{r=1}^{n} \frac{1}{5 \cdot 25^r}$$

$$= \frac{4}{5}\sum_{r=1}^{n} \left(\frac{1}{25}\right)^r$$

$$= \frac{4}{5} \cdot \frac{\frac{1}{25}\left[1-\left(\frac{1}{25}\right)^n\right]}{1-\frac{1}{25}}$$

$$= \frac{4}{5} \cdot \frac{1}{24}\left[1-\left(\frac{1}{25}\right)^n\right]$$

$$= \frac{1}{30}\left(1-\frac{1}{25^n}\right)$$

Note

To use the $\sum a^r$ formula, the power must be exactly r, not $r+1$, not $2r$. Rewrite first.

6.5 Substitution

$$\sum_{r=m}^{n} \mathrm{f}(r) = \sum_{r=m-k}^{n-k} \mathrm{f}(r+k) \quad \text{(replace } r \text{ with } r+k\text{)}$$

Example 6.7

Given $\sum_{r=2}^{n} \frac{1}{r(r+2)} = \frac{1}{2}\left(\frac{5}{6} - \frac{1}{n+1} - \frac{1}{n+2}\right)$, find $\sum_{r=2}^{n} \frac{1}{r^2-1}$.

$$\begin{aligned}
\sum_{r=2}^{n} \frac{1}{r^2-1} &= \sum_{r=2}^{n} \frac{1}{(r-1)(r+1)} \\
&= \sum_{r=1}^{n-1} \frac{1}{r(r+2)} \quad \text{(replace } r \text{ with } r+1\text{)} \\
&= \sum_{r=1}^{1} \frac{1}{r(r+2)} + \sum_{r=2}^{n-1} \frac{1}{r(r+2)} \\
&= \frac{1}{3} + \frac{1}{2}\left(\frac{5}{6} - \frac{1}{n} - \frac{1}{n+1}\right) \\
&= \frac{3}{4} - \frac{1}{2n} - \frac{1}{2(n+1)}
\end{aligned}$$

Note

Substitution only allows **translation** of r values, e.g. replace r with ..., $r-2$, $r-1$, $r+1$, $r+2$, ...

Scaling is not allowed. E.g. replacing r with $2r$ is wrong. List out the terms to see why.

6.6 Convergence

Example 6.8

Given $u_n = \frac{1}{n(n+1)}$ and $S_n = \sum_{r=1}^{n} u_r = 1 - \frac{1}{n+1}$.

Convergence of u_n: let $n \to \infty$ in u_n.	**Convergence of S_n:** let $n \to \infty$ in S_n.
$u_n = \frac{1}{n(n+1)}$ As $n \to \infty,\ u_n \to 0$	$S_n = \sum_{r=1}^{n} \frac{1}{r(r+1)}$ $= 1 - \frac{1}{n+1}$ As $n \to \infty,\ S_n \to 1$ (finite) $\therefore$ the series converges, $S_\infty = 1$

Note

(1) $u_n \to 0$ is not strong enough evidence that S_n converges, e.g. $1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \cdots = \infty$.
(2) However, if S_n converges, then u_n **must** converge to 0.

6.7 Finding u_n or S_n from Each Other

6.7.1 Finding u_n from S_n

$$u_n = S_n - S_{n-1}, \quad u_1 = S_1$$

Example 6.9

$$\begin{aligned} S_n &= n^2 + 2n \\ u_n &= S_n - S_{n-1} \\ &= (n^2 + 2n) - [(n-1)^2 + 2(n-1)] \\ &= 2n + 1 \end{aligned}$$

FLAG N14 [MINOR] — for Maximus to review: 6.7.1 Example: $u_n = S_n - S_{n-1}$ is valid for $n \geqslant 2$ only. The example never states "for $n \geqslant 2$ or checks $n = 1$ ($u_1 = S_1 = 3$ and $2(1) + 1 = 3$, so the formula happens to hold). Add the check; without it the answer is incomplete.

6.7.2 Finding S_n from u_n

Example 6.10

$$u_r = 3r^2 + r$$

$$S_n = \sum_{r=1}^{n} (3r^2 + r)$$

$$= 3\sum_{r=1}^{n} r^2 + \sum_{r=1}^{n} r$$

$$= 3 \cdot \frac{n(n+1)(2n+1)}{6} + \frac{n(n+1)}{2}$$

FLAG N15 [MINOR] — for Maximus to review: 6.7.2 Example: the working stops at an unsimplified expression. Simplifying gives $S_n = \frac{n(n+1)(2n+1)}{2} + \frac{n(n+1)}{2} = n(n+1)^2$. Finish the example (the point of the example is the final closed form).

7 Arithmetic and Geometric Progressions

An **arithmetic progression (AP)** is a sequence with a common difference d, e.g. 2, 5, 8, 11, …

A **geometric progression (GP)** is a sequence with a common ratio r, e.g. 3, 6, 12, 24, …

u_n: the nth term, S_n: the sum of the first term up to the nth term.

	AP	**GP**
Property	common difference d	common ratio r
Example	2, 5, 8, 11, … $(d=3)$	3, 6, 12, 24, … $(r=2)$
u_n	$u_n=a+(n-1)d$	$u_n=ar^{n-1}$
S_n	$S_n=\frac{n}{2}[2a+(n-1)d]$ $S_n=\frac{n}{2}(a+l)$	$S_n=\frac{a(1-r^n)}{1-r}=\frac{a(r^n-1)}{r-1}, r\neq 1$
S_∞	not convergent	$S_\infty=\frac{a}{1-r}$, only if $\lvert r\rvert<1$
Proving	$u_{n+1}-u_n=d$ (constant) $u_3-u_2=k_1, u_2-u_1=k_2$ $k_1\neq k_2 \therefore$ not an AP	$\frac{u_{n+1}}{u_n}=r$ (constant) $\frac{u_3}{u_2}=k_1, \frac{u_2}{u_1}=k_2$ $k_1\neq k_2 \therefore$ not a GP

FLAG N16 [STATEMENT] — for Maximus to review: Ch 7 table, row S_∞ for an AP: "not convergent is true only when a and d are not both 0 (an AP with $a=d=0$ has $S_n=0$ for all n, so $S_\infty=0$). Say "diverges unless $a=d=0$ (or "does not converge for a non-trivial AP).

However, to prove that something IS an AP, you have to do it in terms of n. You cannot compare u_1, u_2 and u_3 in the same way that we prove that something is not an AP.

Note: AP: u_2, u_4, u_6, … is an AP (common difference $2d$), GP: u_2, u_4, u_6, … is a GP (common ratio r^2).

> **Important Note**
>
> Remember that you can use a GC table whenever needed, especially when the variable you are solving for is an integer. It is very common in this chapter because a lot of the time you are solving for $n\in\mathbb{Z}^+$.

7.1 Summing

Example 7.1

AP: first term 9, common difference 2. Sum of the 20th to 50th terms inclusive:

$$\begin{aligned} S_{50}-S_{19} &= \frac{50}{2}[2(9)+49(2)]-\frac{19}{2}[2(9)+18(2)] \\ &= 2900-513 \\ &= 2387 \end{aligned}$$

7.2 Common Questions

7.2.1 Comparing AP and GP terms

Example 7.2

[TMJC 2025 Prelim P1 Q4]

An arithmetic series has first term a and common difference d, where a and d are non-zero real constants. A convergent geometric series has first term b and common ratio r, where $b>0$ and r is a non-zero real constant. The sum of the first n terms of the geometric series is denoted by S.

It is given that the 7th, 10th and 11th terms of the arithmetic series are equal to the 3rd, 6th and 10th terms of the geometric series respectively.

$$a+6d=br^2 \quad -- (1)$$
$$a+9d=br^5 \quad -- (2)$$
$$a+10d=br^9 \quad -- (3)$$

7.2.2 Bank Questions

Important Note

Any bank questions will come in the form $u_{n+1}=pu_n+q$, where p and q are constants.

FLAG N17 [MINOR] — for Maximus to review: 7.2.2 Bank Questions, info box: "Any bank questions will come in the form $u_{n+1}=pu_n+q$ is over-general (no bank/loan question is guaranteed to take that form, and the example below is solved by summing a GP rather than via $u_{n+1}=pu_n+q$). Soften to "Many/typical bank questions….

Example 7.3

[TJC 2025 Prelim P1 Q7]

Janice, on the other hand, deposits \$100 at the start of every month into a bank account that earns 0.3% interest per month, with interest calculated and added into the account at the end of each month.

Find the amount that Janice will have in her savings account at the end of N months.

FLAG N18 [MINOR] — for Maximus to review: 7.2.2 Bank Questions, example: the extract begins "Janice, *on the other hand*,…but Mabel's part of the question is not shown, so "on the other hand has nothing to contrast with. Quote the Mabel sentence or drop the phrase.

nth mth	Amount at start of month ($)	Amount at end of month ($)
1 Jan26	100	$100(1.003)$
2	$100+100(1.003)$ $=100(1+1.003)$	$100(1+1.003)(1.003)$ $=100\,(1.003+1.003^2)$
3	$100+100\,(1.003+1.003^2)$ $=100\,(1+1.003+1.003^2)$	$100\,(1+1.003+1.003^2)\,(1.003)$ $=100\,(1.003+1.003^2+1.003^3)$
⋮	⋮	⋮
N		$100\,(1.003+1.003^2+\cdots+1.003^N)$ $=\dfrac{100(1.003)\,(1.003^N-1)}{1.003-1}$

7.2.3 Contextual Questions

For general contextual questions, it is important to determine whether the question is asking for u_n or S_n.

Example 7.4

[RVHS 2025 Prelim P2 Q4]

The dosage of medicine given to a patient needs to be carefully managed in order to achieve the desired result.

(a) To give the patient time to adapt to a particular medicine, the dosage is slowly increased over time. The patient is initially given a dosage of 0.75 units. The dosage is increased by 0.5 units each day until a total of at least 10 units of medicine has been taken. Find the least number of days it takes to achieve this, and the dosage on that day. [4]

(b) The doctor prescribed a new medication to another patient. After n days, the total dosage of the new medicine administered to the patient is given by $\dfrac{3}{20}\left(\dfrac{5^n-4^n}{5^{n-1}}\right)$ units.

(i) Show that the increase in dosage of the new medicine administered per day follows a geometric progression. [3]

(ii) Find the total dosage of the new medicine administered to the patient after a long period of treatment. [2]

(c) Doctors can manage the medication so that the daily dosage of medicine increases as an arithmetic progression. Explain in context why the arithmetic progression is not a preferred model for the dosage of medication. [1]

8 Differentiation

8.1 Basic Differentials

y	$\frac{dy}{dx}$
x^n	nx^{n-1}
$\sin x$	$\cos x$
$\cos x$	$-\sin x$
$\tan x$	$\sec^2 x$
$\log_b x$	$\frac{1}{x \ln b}$, b constant
$e^{f(x)}$	$f'(x)\, e^{f(x)}$
$a^{f(x)}$	$f'(x)\, a^{f(x)} \ln a$, a constant
$p(x)\, q(x)$	$p'(x)\, q(x) + p(x)\, q'(x)$
$\frac{p(x)}{q(x)}$	$\frac{p'(x)\, q(x) - p(x)\, q'(x)}{[q(x)]^2}$

y	$\frac{dy}{dx}$
$\sin^{-1} x$	$\frac{1}{\sqrt{1-x^2}}$
$\cos^{-1} x$	$-\frac{1}{\sqrt{1-x^2}}$
$\tan^{-1} x$	$\frac{1}{1+x^2}$
$\sec x$	$\sec x \tan x$
$\operatorname{cosec} x$	$-\operatorname{cosec} x \cot x$
$\cot x$	$-\operatorname{cosec}^2 x$

Important Note

Remember to multiply by the differential when you are differentiating. Bracket, not divide.

8.1.1 Implicit Differentiation

Example 8.1

$$x^3y+1+\ln y=\cos(xy)$$

Differentiate both sides with respect to x:

$$3x^2y+x^3\frac{\mathrm{d}y}{\mathrm{d}x}+0+\frac{1}{y}\frac{\mathrm{d}y}{\mathrm{d}x}=-\sin(xy)\left(y+x\frac{\mathrm{d}y}{\mathrm{d}x}\right)$$

$$3x^2y+x^3\frac{\mathrm{d}y}{\mathrm{d}x}+\frac{1}{y}\frac{\mathrm{d}y}{\mathrm{d}x}=-y\sin(xy)-x\sin(xy)\frac{\mathrm{d}y}{\mathrm{d}x}$$

$$x^3\frac{\mathrm{d}y}{\mathrm{d}x}+\frac{1}{y}\frac{\mathrm{d}y}{\mathrm{d}x}+x\sin(xy)\frac{\mathrm{d}y}{\mathrm{d}x}=-y\sin(xy)-3x^2y$$

$$\frac{\mathrm{d}y}{\mathrm{d}x}\left(x^3+\frac{1}{y}+x\sin(xy)\right)=-y\left(\sin(xy)+3x^2\right)$$

$$\frac{\mathrm{d}y}{\mathrm{d}x}\left(\frac{x^3y+1+xy\sin(xy)}{y}\right)=-y\left(\sin(xy)+3x^2\right)$$

$$\therefore\ \frac{\mathrm{d}y}{\mathrm{d}x}=-\frac{y^2\left(\sin(xy)+3x^2\right)}{x^3y+1+xy\sin(xy)}$$

8.1.2 Parametric Differentiation

$$\frac{\mathrm{d}y}{\mathrm{d}x}=\frac{\mathrm{d}y}{\mathrm{d}t}\times\frac{\mathrm{d}t}{\mathrm{d}x}$$

Example 8.2

$$x=t+\sin t,\quad y=1-\cos t$$

$$\frac{\mathrm{d}x}{\mathrm{d}t}=1+\cos t,\quad \frac{\mathrm{d}y}{\mathrm{d}t}=\sin t$$

$$\frac{\mathrm{d}t}{\mathrm{d}x}=\frac{1}{\frac{\mathrm{d}x}{\mathrm{d}t}}=\frac{1}{1+\cos t}$$

$$\frac{\mathrm{d}y}{\mathrm{d}x}=\frac{\mathrm{d}y}{\mathrm{d}t}\times\frac{\mathrm{d}t}{\mathrm{d}x}$$

$$=\frac{\sin t}{1+\cos t}$$

$$=\frac{2\sin\frac{t}{2}\cos\frac{t}{2}}{2\cos^2\frac{t}{2}}$$

$$=\frac{\sin\frac{t}{2}}{\cos\frac{t}{2}}$$

$$\therefore\ \frac{\mathrm{d}y}{\mathrm{d}x}=\tan\frac{t}{2}$$

9 Applications of Differentiation

9.1 Tangents & Normals

A tangent can be diagonal, vertical or horizontal.

Gradient of tangent = value of $\frac{dy}{dx}$.

Gradient of normal:

$$m_{\text{tangent}} \times m_{\text{normal}} = -1$$

Equation of tangent: $y = mx + c$. To find c, substitute any point on the tangent.

FLAG N19 [STATEMENT] — for Maximus to review: 9.1 Tangents & Normals: the opening line says a tangent can be diagonal, vertical or horizontal, but $m_{\text{tangent}} \times m_{\text{normal}} = -1$ and "$y = mx + c$ do not cover those cases. Horizontal tangent ($m = 0$): the normal is vertical, $x = k$. Vertical tangent ($\frac{dy}{dx}$ undefined): the tangent is $x = k$ and the normal is horizontal, $y = k$. Add this case.

9.1.1 Standard Curves

Example 9.1

Find the equation of the tangent to $y = x^2 - 3x + 5$ at $x = 2$.

When $x = 2$, $y = 2^2 - 3(2) + 5 = 3$

$$\frac{dy}{dx} = 2x - 3$$

When $x = 2$, $\frac{dy}{dx} = 2(2) - 3 = 1$

$$y = x + c$$

$$3 = 2 + c$$

$$c = 1$$

$$\therefore y = x + 1$$

9.1.2 Implicit Curves

Example 9.2

[EJC 2025 Prelim P2 Q1(b)]

The curve C has equation $x^3 + y^3 - 3xy = k$, where k is a positive constant. There are two points on C where the tangents are parallel to the y-axis. Given that $k = 3$, find the exact y-coordinates of these two points.

$$\frac{dy}{dx} = \frac{y - x^2}{y^2 - x}$$

Tangent parallel to y-axis $\Rightarrow \frac{dy}{dx}$ undefined:

$$y^2 - x = 0$$
$$x = y^2$$
$$y^6 - 2y^3 - 3 = 0$$
$$(y^3 - 3)(y^3 + 1) = 0$$
$$\therefore y = 3^{\frac{1}{3}} \quad \text{or} \quad y = -1$$

9.1.3 Parametric Curves

Example 9.3

[MI 2025 Prelim P2 Q2(i)]

A curve C has parametric equations $x = \sqrt{t}$, $y = t^2 + 4$, where $0 \leqslant t \leqslant 16$. Find the equation of the tangent to C at the point P with parameter p.

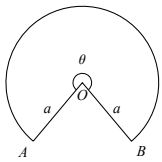
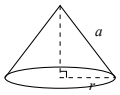
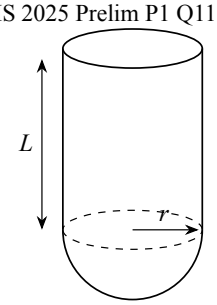
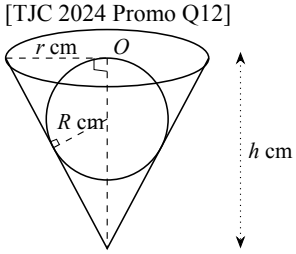
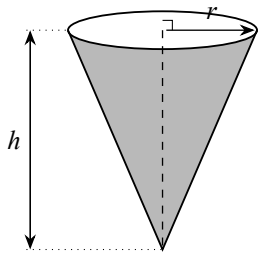
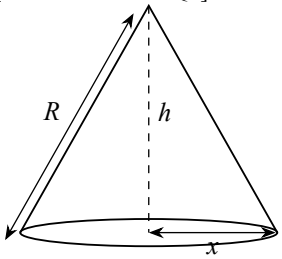
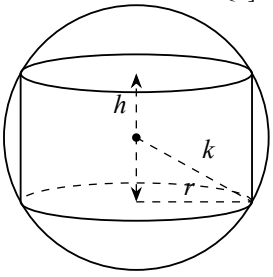
$$\frac{dx}{dt} = \frac{1}{2\sqrt{t}}, \quad \frac{dy}{dt} = 2t$$
$$\frac{dy}{dx} = \frac{2t}{\frac{1}{2\sqrt{t}}} = 4t^{\frac{3}{2}}$$
$$y - (p^2 + 4) = 4p^{\frac{3}{2}}\left(x - \sqrt{p}\right)$$
$$\therefore y = 4p^{\frac{3}{2}}x - 3p^2 + 4$$

9.2 Max/Min

- To find the stationary value of y, solve $\frac{dy}{dx} = 0$.
- Then, to find/prove its nature, perform the first/second derivative test.

Important Note

In many questions, you may be asked to prove the equation of the quantity you are trying to maximise. It's important to not spend too long on it if you can't prove it. Just continue with the rest of the question to differentiate and solve $\frac{dy}{dx} = 0$.

Index	Diagram	Info	Equation	Comments
1	[TJC 2025 Prelim P2 Q4]  Figure 1  Figure 2	Constant: radius a Variables: angle θ	Prove that $V^2 = \frac{a^6}{576\pi^4}\theta^4\left(4\pi^2 - \theta^2\right)$	Take advantage of the constant slant height and major arc length AB.
2	[DHS 2025 Prelim P1 Q11]  	Constant: budget $\$k$ Variables: r	Prove that $V = \frac{k}{20}r - \frac{10}{3}\pi r^3$	Use the constant budget k to eliminate L.
3	[TJC 2024 Promo Q12]  	Constant: sphere radius R Variables: h	Prove that $r^2 = \frac{h^2R^2}{h^2 - 2hR}$	Take advantage of the constant sphere radius R, and similar triangles. Pythagoras theorem is useful here too.
4	[CJC 2024 Promo Q10]  	Constant: volume $20\pi\ \text{cm}^3$ Variables: r	Prove that $A\frac{\mathrm{d}A}{\mathrm{d}r} = \pi^2\left(2r^3 - \frac{3600}{r^3}\right)$	Use the constant volume to eliminate h. For rate of change questions, use similar triangles.
5	[EJC 2018 Promo Q7]  	Constant: slant edge R Variables: x	Prove that $9V^2 = \pi^2x^4\left(R^2 - x^2\right)$	Use the constant R to eliminate h.
6	[TMJC 2024 Prelim P1 Q2]  	Constant: sphere radius k Variables: k, r	—	Use the constant radius of sphere k and Pythagoras theorem is useful here too.

FLAG N20 [MINOR] — for Maximus to review: 9.2 Max/Min table, row 6 (TMJC 2024 Prelim P1 Q2): "Variables: k, r — k is the constant (sphere radius, as the "Constant cell says directly above), so the only variable is r. Compare rows 1–5. Also the "Equation cell of row 6 is "—.

9.3 Rate of Change

$$\frac{dy}{dt} = \frac{dy}{dx} \times \frac{dx}{dt}$$

Step 1: Identify what information is given.

Step 2: Form the missing equation if necessary and differentiate it.

Example 9.4

Given $y = x^2 - 4x + 1$ and $\frac{dx}{dt} = 3$, find the rate of change of y when $x = 5$.

$$\frac{dy}{dx} = 2x - 4$$

When $x = 5$, $\frac{dy}{dx} = 2(5) - 4 = 6$

$$\frac{dy}{dt} = \frac{dy}{dx} \times \frac{dx}{dt}$$

$$= 6 \times 3$$

$$\therefore \frac{dy}{dt} = 18$$

10 Power Series

10.1 Standard Formulae (MF27)

$$(1+x)^n = 1+nx+\frac{n(n-1)}{2!}x^2+\cdots+\frac{n(n-1)\cdots(n-r+1)}{r!}x^r+\ldots \quad (|x|<1) \tag{1}$$

$$e^x = 1+x+\frac{x^2}{2!}+\cdots+\frac{x^r}{r!}+\ldots \quad (\text{all } x) \tag{2}$$

$$\sin x = x-\frac{x^3}{3!}+\frac{x^5}{5!}-\cdots+\frac{(-1)^r x^{2r+1}}{(2r+1)!}+\ldots \quad (\text{all } x) \tag{3}$$

$$\cos x = 1-\frac{x^2}{2!}+\frac{x^4}{4!}-\cdots+\frac{(-1)^r x^{2r}}{(2r)!}+\ldots \quad (\text{all } x) \tag{4}$$

$$\ln(1+x) = x-\frac{x^2}{2}+\frac{x^3}{3}-\cdots+\frac{(-1)^{r+1}x^r}{r}+\ldots \quad (-1<x\leqslant 1) \tag{5}$$

Important Note

For equations (1) and (5), the constant must be 1.

CAUTION

Any substituted expressions of x cannot have + constant inside.

✗ $e^{1+x+x^2+\cdots} = 1+(1+x+x^2+\cdots)+\frac{(1+x+x^2+\cdots)^2}{2!}+\cdots$

✓ $e^{1+x+x^2+\cdots} = e^1\cdot e^{x+x^2+\cdots}$

$$= e\left[1+(x+x^2+\cdots)+\frac{(x+x^2+\cdots)^2}{2!}+\cdots\right]$$

FLAG N21 [STATEMENT] — for Maximus to review: 10.1 Standard Formulae, caution box: "cannot have + constant inside is imprecise. The line marked ✗ is a valid identity (formula (2) holds for every argument); it just does not give a series in ascending powers of x. The real rule: to expand in powers of x, the substituted expression must tend to 0 as $x\to 0$ for (2)–(4), and for (1) and (5) the constant must be 1 (factor it out first). Reword so the ✗ is not called "wrong.

10.2 Maclaurin Series

$$f(x) = f(0)+x\,f'(0)+\frac{x^2}{2!}f''(0)+\cdots+\frac{x^r}{r!}f^{(r)}(0)+\ldots$$

Conditions for existence: for all $n\in\mathbb{Z}^+\cup\{0\}$, $f^{(n)}$ is differentiable and $f^{(n)}(0)$ is defined.

10.3 Small Angle Approximations

When θ is sufficiently small, θ^3 and above can be ignored.

$$\sin\theta \approx \theta$$

$$\cos\theta \approx 1-\frac{\theta^2}{2}$$

$$\tan\theta \approx \theta$$

Example 10.1

$$\sin 2\theta \approx 2\theta$$

Example 10.2

$$\begin{aligned}\cos 3\theta &\approx 1-\frac{(3\theta)^2}{2}\\ &= 1-\frac{9\theta^2}{2}\end{aligned}$$

Example 10.3

$$\begin{aligned}\sec\theta &= \frac{1}{\cos\theta}\\ &\approx \left(1-\frac{\theta^2}{2}\right)^{-1}\\ &\approx 1+\frac{\theta^2}{2}\end{aligned}$$

Example 10.4

$\sin\left(\frac{\pi}{4}+\theta\right) \neq \frac{\pi}{4}+\theta$

$$\begin{aligned}\sin\left(\frac{\pi}{4}+\theta\right) &= \sin\frac{\pi}{4}\cos\theta+\cos\frac{\pi}{4}\sin\theta\\ &= \frac{1}{\sqrt{2}}(\cos\theta+\sin\theta)\\ &\approx \frac{1}{\sqrt{2}}\left(1-\frac{\theta^2}{2}+\theta\right)\\ \therefore\ \sin\left(\frac{\pi}{4}+\theta\right) &\approx \frac{1}{\sqrt{2}}\left(1+\theta-\frac{\theta^2}{2}\right)\end{aligned}$$

10.4 General Approximations

- Values of x closer to 0 give better approximations as higher powers of x are less significant.
- Values of x outside the range of validity give an inaccurate approximation.

11 Integration

11.1 Basic Integrals

Basic	**General form** (let a, b, k be constants)
$\int x^n \,\mathrm{d}x = \frac{x^{n+1}}{n+1} + c, \quad n \neq -1$	$\int (ax+b)^n \,\mathrm{d}x = \frac{(ax+b)^{n+1}}{a(n+1)} + c, \quad n \neq -1$
$\int \frac{1}{x} \,\mathrm{d}x = \ln\lvert x\rvert + c$	$\int \frac{1}{ax+b} \,\mathrm{d}x = \frac{1}{a}\ln\lvert ax+b\rvert + c$
$\int \mathrm{e}^x \,\mathrm{d}x = \mathrm{e}^x + c$	$\int \mathrm{e}^{kx} \,\mathrm{d}x = \frac{1}{k}\mathrm{e}^{kx} + c$
$\int \sin x \,\mathrm{d}x = -\cos x + c$	$\int \sin kx \,\mathrm{d}x = -\frac{1}{k}\cos kx + c$
$\int \cos x \,\mathrm{d}x = \sin x + c$	$\int \cos kx \,\mathrm{d}x = \frac{1}{k}\sin kx + c$
$\int \sec^2 x \,\mathrm{d}x = \tan x + c$	$\int \sec^2 kx \,\mathrm{d}x = \frac{1}{k}\tan kx + c$

11.2 $\mathrm{f}'(x)$ Formulas

$$\int \mathrm{f}'(x)\,[\mathrm{f}(x)]^n \,\mathrm{d}x = \frac{[\mathrm{f}(x)]^{n+1}}{n+1} + c, \qquad n \neq -1$$

$$\int \frac{\mathrm{f}'(x)}{\mathrm{f}(x)} \,\mathrm{d}x = \ln\lvert \mathrm{f}(x)\rvert + c$$

$$\int \mathrm{f}'(x)\,\mathrm{e}^{\mathrm{f}(x)} \,\mathrm{d}x = \mathrm{e}^{\mathrm{f}(x)} + c$$

11.3 New Trigonometric Integrals

- To find $\int \tan x \,\mathrm{d}x$, $\int \sec x \,\mathrm{d}x$, $\int \operatorname{cosec} x \,\mathrm{d}x$, $\int \cot x \,\mathrm{d}x$ — see the **MF27**.
- To find $\int \sin^2 x \,\mathrm{d}x$ or $\int \cos^2 x \,\mathrm{d}x$, use the **double angle formula** for $\cos 2\theta$.
  To find $\int \tan^2 x \,\mathrm{d}x$, use the identity $\tan^2 x = \sec^2 x - 1$.
- The differentiation formulas for $\sec x$ and $\operatorname{cosec} x$ in the MF27 can also be used as **integration** formulas. In particular, the missing formula yields

$$\int \operatorname{cosec}^2 x \,\mathrm{d}x = -\cot x + c.$$

FLAG N22 [MINOR] — for Maximus to review: 11.3 New Trigonometric Integrals: no mention of products such as $\sin mx \cos nx$ (also $\sin mx \sin nx$, $\cos mx \cos nx$), which are integrated by first using the factor formulae from the MF27. These appear in H2 integration questions; consider adding a bullet.

11.4 Standard MF27 Formulas

Standard	General form
$\int \frac{1}{a^2+x^2}\,dx = \frac{1}{a}\tan^{-1}\frac{x}{a}+c$	$\int \frac{f'(x)}{a^2+[f(x)]^2}\,dx = \frac{1}{a}\tan^{-1}\frac{f(x)}{a}+c$
$\int \frac{1}{\sqrt{a^2-x^2}}\,dx = \sin^{-1}\frac{x}{a}+c$	$\int \frac{f'(x)}{\sqrt{a^2-[f(x)]^2}}\,dx = \sin^{-1}\frac{f(x)}{a}+c$
$\int \frac{1}{x^2-a^2}\,dx = \frac{1}{2a}\ln\left\lvert\frac{x-a}{x+a}\right\rvert+c$	$\int \frac{f'(x)}{[f(x)]^2-a^2}\,dx = \frac{1}{2a}\ln\left\lvert\frac{f(x)-a}{f(x)+a}\right\rvert+c$
$\int \frac{1}{a^2-x^2}\,dx = \frac{1}{2a}\ln\left\lvert\frac{a+x}{a-x}\right\rvert+c$	$\int \frac{f'(x)}{a^2-[f(x)]^2}\,dx = \frac{1}{2a}\ln\left\lvert\frac{a+f(x)}{a-f(x)}\right\rvert+c$

IMPORTANT

a must be a (positive) **constant**.

Important Remark

When using these formulas, it is **VERY COMMON** to have to complete the square first.

Remark

These formulas are commonly used together with $f'(x)$ formulas to integrate **rational functions**.

Example 11.1

$$\begin{aligned}\int \frac{4x}{\sqrt{4x-x^2}}\,dx &= \int \frac{-2(-2x+4)+8}{\sqrt{4x-x^2}}\,dx \\ &= -2\int (-2x+4)\,(4x-x^2)^{-\frac{1}{2}}\,dx + 8\int \frac{1}{\sqrt{4-(x-2)^2}}\,dx \\ &= -2\cdot\frac{(4x-x^2)^{\frac{1}{2}}}{\left(\frac{1}{2}\right)} + 8\sin^{-1}\frac{x-2}{2}+c \\ &= -4\sqrt{4x-x^2}+8\sin^{-1}\frac{x-2}{2}+c\end{aligned}$$

11.5 Integration by Parts (IKID)

Integrate Keep Integrate Differentiate

$$IK - \int ID \, \mathrm{d}x$$

Example 11.2

$$\int \underbrace{x}_{\text{K/D}} \, \underbrace{\mathrm{e}^x}_{\text{I}} \, \mathrm{d}x = \underbrace{\mathrm{e}^x}_{\text{I}} \, \underbrace{x}_{\text{K}} - \int \underbrace{\mathrm{e}^x}_{\text{I}} \, \underbrace{1}_{\text{D}} \, \mathrm{d}x$$

- It is possible to make the **wrong choice** of which expression to integrate. In such scenarios, the $\int ID \, \mathrm{d}x$ will look *worse* than the original problem.
- It is common to have to **IKID twice**. Consider this approach if you are unable to continue.

> **IMPORTANT**
>
> Integration by parts is often used as the **last resort**, i.e. please consider **ALL OTHER INTEGRATION TECHNIQUES** first.

Common tricks

- **IKID by multiplying by 1.**

$$\int \ln x \, \mathrm{d}x = \int 1 \cdot \ln x \, \mathrm{d}x = \ldots$$

- **Non-obvious splitting.**

$$\int x^3 \mathrm{e}^{x^2} \, \mathrm{d}x = \int x^2 \cdot x\mathrm{e}^{x^2} \, \mathrm{d}x = \ldots$$

  Here, we can integrate $x\mathrm{e}^{x^2}$ using the $\int \mathrm{f}'(x)\mathrm{e}^{\mathrm{f}(x)} \, \mathrm{d}x$ formula.

- **Loop method.**

$$\begin{aligned}\int \mathrm{e}^x \cos x \, \mathrm{d}x &= \mathrm{e}^x \cos x + \int \mathrm{e}^x \sin x \, \mathrm{d}x \\ &= \mathrm{e}^x \cos x + \mathrm{e}^x \sin x - \int \mathrm{e}^x \cos x \, \mathrm{d}x\end{aligned}$$

  Therefore,

$$\begin{aligned}2\int \mathrm{e}^x \cos x \, \mathrm{d}x &= \mathrm{e}^x \cos x + \mathrm{e}^x \sin x + c \\ \int \mathrm{e}^x \cos x \, \mathrm{d}x &= \frac{1}{2}(\mathrm{e}^x \cos x + \mathrm{e}^x \sin x) + c\end{aligned}$$

11.6 Integration by Substitution

Given $\int f(x)\,dx$ and a substitution $u=\ldots$, apply:

Step 1: Substitute u into $f(x)$.

Step 2: Find $\frac{dx}{du}$, and substitute it into the integral.

Step 3: *(Only for definite integrals)* Substitute the limits of integration from x values to u values.

Indefinite Integral

Example 11.3

$$\int \sqrt{1-x^2}\,dx, \qquad x=\sin u$$

$$= \int \underbrace{\sqrt{1-\sin^2 u}}_{\text{Step 1}}\ \underbrace{\cos u\,du}_{\text{Step 2}}$$

$$= \int \sqrt{\cos^2 u}\,\cos u\,du$$

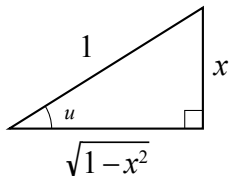
$$= \int \cos^2 u\,du$$

$$= \frac{1}{2}\int (1+\cos 2u)\,du$$

$$= \frac{1}{2}\left(u+\frac{1}{2}\sin 2u\right)+c$$

$$= \frac{1}{2}u+\frac{1}{2}\sin u\cos u+c$$

$$= \frac{1}{2}\sin^{-1}x+\frac{1}{2}x\sqrt{1-x^2}+c$$



$\sin u = x$

$\cos u = \sqrt{1-x^2}$

Definite Integral

Example 11.4

$$\int_0^{\frac{1}{2}} \sqrt{1-x^2}\,dx, \qquad x=\sin u$$

$$x=0 \Rightarrow u=0$$

$$x=\frac{1}{2} \Rightarrow u=\frac{\pi}{6}$$

$$\int_0^{\frac{1}{2}} \sqrt{1-x^2}\,dx = \int_0^{\frac{\pi}{6}} \sqrt{1-\sin^2 u}\,\cos u\,du$$

$$= \int_0^{\frac{\pi}{6}} \cos^2 u\,du$$

$$= \frac{1}{2}\int_0^{\frac{\pi}{6}} (1+\cos 2u)\,du$$

$$= \frac{1}{2}\left[u+\frac{1}{2}\sin 2u\right]_0^{\frac{\pi}{6}}$$

$$= \frac{1}{2}\left[\left(\frac{\pi}{6}+\frac{1}{2}\sin\frac{\pi}{3}\right)-\left(0+\frac{1}{2}\sin 0\right)\right]$$

$$= \frac{1}{2}\left(\frac{\pi}{6}+\frac{\sqrt{3}}{4}\right)$$

$$= \frac{\pi}{12}+\frac{\sqrt{3}}{8}$$

FLAG N23 [MINOR] — for Maximus to review: 11.6 Integration by Substitution (header vs example): the header says "a substitution $u=\ldots$ and Step 2 says "find $\frac{dx}{du}$, but both worked examples use $x=\sin u$ (i.e. x in terms of u). State both forms ($u=g(x) \Rightarrow du=g'(x)\,dx$, and $x=h(u) \Rightarrow dx=h'(u)\,du$) or make the header match the examples.

FLAG N24 [MINOR] — for Maximus to review: 11.6 Example (indefinite and definite): $\sqrt{\cos^2 u}=|\cos u|$, not $\cos u$ in general. It equals $\cos u$ here only because $u\in[-\frac{\pi}{2},\frac{\pi}{2}]$ (from $x=\sin u$). Say so (or write $|\cos u|=\cos u$ since $\cos u \geqslant 0$).

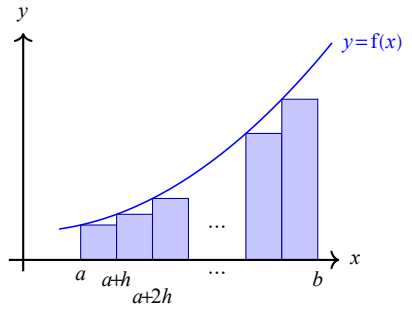
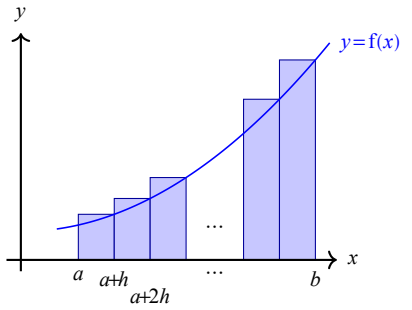
Remark

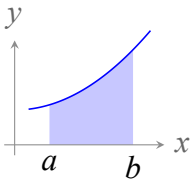
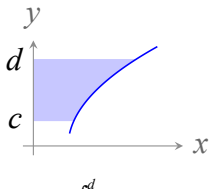
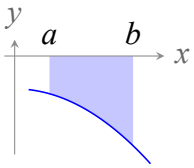
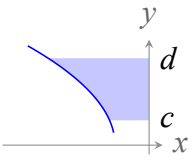
Remember to sub the final answer back to be in terms of x, unless it's a **definite integral**.

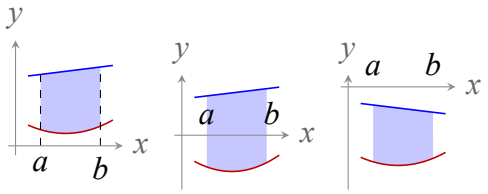
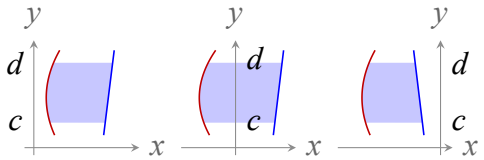
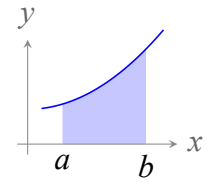
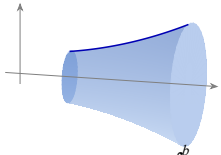
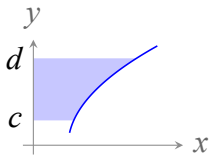
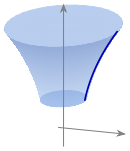
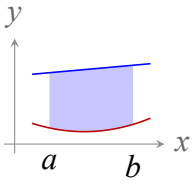
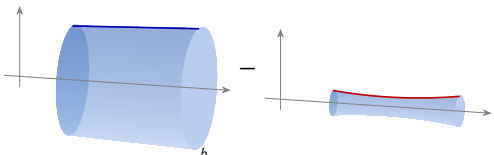
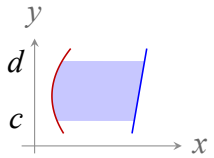
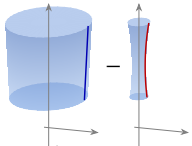
12 Definite Integrals

$$\int_a^b \mathrm{f}(x)\,\mathrm{d}x = \Big[\mathrm{F}(x)\Big]_a^b = \mathrm{F}(b) - \mathrm{F}(a)$$

12.1 Area as a Limit of a Sum

Left-hand sum	**Right-hand sum**
	
$\lim\limits_{n\to\infty}\sum\limits_{r=0}^{n-1}\dfrac{b-a}{n}\,\mathrm{f}\left(a+\dfrac{r(b-a)}{n}\right)=\int_a^b \mathrm{f}(x)\,\mathrm{d}x$	$\lim\limits_{n\to\infty}\sum\limits_{r=1}^{n}\dfrac{b-a}{n}\,\mathrm{f}\left(a+\dfrac{r(b-a)}{n}\right)=\int_a^b \mathrm{f}(x)\,\mathrm{d}x$
Increasing $\mathrm{f}(x) \Rightarrow$ underestimate Decreasing $\mathrm{f}(x) \Rightarrow$ overestimate	Increasing $\mathrm{f}(x) \Rightarrow$ overestimate Decreasing $\mathrm{f}(x) \Rightarrow$ underestimate

Index	d***x***	d***y***
Area under curve	 $\int_a^b y\,\mathrm{d}x$	 $\int_c^d x\,\mathrm{d}y$
Negative areas	 $-\int_a^b y\,\mathrm{d}x$	 $-\int_c^d x\,\mathrm{d}y$

Index	dx	dy
Areas between two graphs	 All 3 cases: $\int_a^b (y_1 - y_2)\ \mathrm{d}x$ y_1 (blue) above y_2 (red)	 All 3 cases: $\int_c^d (x_1 - x_2)\ \mathrm{d}y$ x_1 (blue) right of x_2 (red)
Volume rotated about axis	  About the x-axis: $\pi\int_a^b y^2\ \mathrm{d}x$	  About the y-axis: $\pi\int_c^d x^2\ \mathrm{d}y$
Volume generated by region between two curves	  $\pi\int_a^b (y_1^2 - y_2^2)\ \mathrm{d}x$	  $\pi\int_c^d (x_1^2 - x_2^2)\ \mathrm{d}y$

When finding **area**, we generally have the choice to integrate via dx or dy. Some considerations:

- dx or dy easier to integrate
- do we need to add or subtract regular shapes like rectangles
- is it even possible to make x (or y) the subject?

FLAG N25 [OMISSION] — for Maximus to review: 12.1 Area as a Limit of a Sum / areas: the area under a parametric curve ($\int y \frac{\mathrm{d}x}{\mathrm{d}t}\,\mathrm{d}t$ with limits converted to t) is in the 9758 syllabus but is not covered in this chapter. Consider adding a short section.

For **volume**, we cannot choose which variable to integrate with respect to.

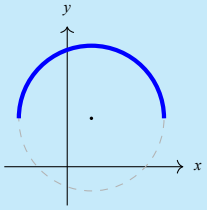
- Region rotated about the x-axis: $\pi\int_a^b y^2\,\mathrm{d}x$
- Region rotated about the y-axis: $\pi\int_c^d x^2\,\mathrm{d}y$

Important Note: dealing with conics

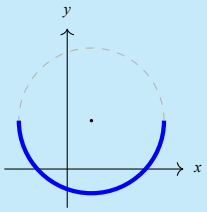
When making y the subject, the sign of the square root indicates which half of the conic is being considered.

Example $(x-1)^2+(y-2)^2=9$

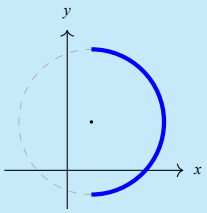
$y=2+\sqrt{9-(x-1)^2}$



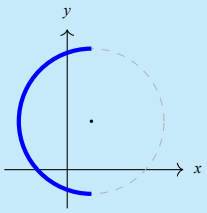
$y=2-\sqrt{9-(x-1)^2}$



$x=1+\sqrt{9-(y-2)^2}$



$x=1-\sqrt{9-(y-2)^2}$



Note: this concept can be used for any conic. That is, circles, ellipses, hyperbolas, parabolas.

Notes on volume

- "Rotate π radians about …" does not usually mean the formula has to be divided by 2. It just means that rotating only 180° is sufficient to generate the full solid. The volume remains as $\pi\int_a^b y^2\,\mathrm{d}x$.
- To find the volume of rotating regions "below" the axis, there is no need to put a negative sign in front:

✗ $-\pi\int_a^b y^2\,\mathrm{d}x$ ✓ $\pi\int_a^b y^2\,\mathrm{d}x$

- Common mistake for volume between two graphs:

✗ $\pi\int_a^b (y_1-y_2)^2\,\mathrm{d}x$ ✓ $\pi\int_a^b (y_1^2-y_2^2)\,\mathrm{d}x$

FLAG N26 [STATEMENT] — for Maximus to review: 12.1 Notes on volume: $\pi\int_a^b (y_1^2-y_2^2)\,\mathrm{d}x$ (here and in the "Volume generated by region between two curves row of the table) is valid only when both curves lie on the same side of the axis of rotation ($y_1 \geqslant y_2 \geqslant 0$, or both $\leqslant 0$). If the region straddles the axis, the outer and inner radii are the larger and smaller of $|y_1|$, $|y_2|$. State the condition.

Important Note

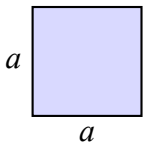
Take note if the question asks for exact or not. If they did not, please evaluate the integral using GC rather than integration by hand.

Remark

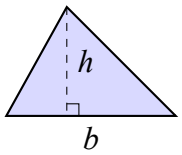
If an integral seems very difficult/impossible to integrate, then check if the question wants exact form OR check if you have integrated something similar in earlier parts.

Regular Shapes in Area and Volume

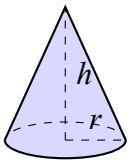
Sometimes we may encounter familiar shapes like squares, rectangles, triangles, trapeziums, cones, and cylinders. It will be useful to use the formulas for these shapes/solids for efficiency.



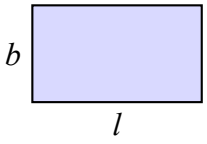
Square: Area $= a^2$



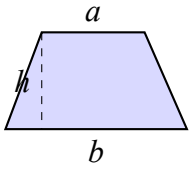
Triangle: Area $= \frac{1}{2}bh$



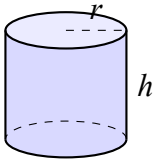
Cone: Volume $= \frac{1}{3}\pi r^2 h$



Rectangle: Area $= lb$



Trapezium: Area $= \frac{1}{2}(a+b)h$



Cylinder: Volume $= \pi r^2 h$

Integration with Modulus

To integrate with modulus, we determine where the function inside the modulus is positive and where it is negative, then split it up.

Example 12.1

$$\begin{aligned}\int_0^{\pi} |x\sin x\cos x|\,\mathrm{d}x &= \int_0^{\frac{\pi}{2}} (x\sin x\cos x)\,\mathrm{d}x + \int_{\frac{\pi}{2}}^{\pi} -(x\sin x\cos x)\,\mathrm{d}x \\ &= \int_0^{\frac{\pi}{2}} \frac{1}{2}x\sin 2x\,\mathrm{d}x + \int_{\frac{\pi}{2}}^{\pi} -\frac{1}{2}x\sin 2x\,\mathrm{d}x \\ &= \ldots\end{aligned}$$

FLAG N27 [MINOR] — for Maximus to review: 12.1.2 Integration with Modulus: the example stops at "=…. Finish it (integrate by parts and give the value) or mark it as an outline.

13 Differential Equations

13.1 Definition of a Differential Equation

Differential Equation

A **differential equation** is an equation involving a derivative, e.g. $\frac{dy}{dx}$, of an unknown function.

13.2 Solving a Differential Equation

Solving a differential equation means finding y as a function of x that satisfies the differential equation.

General and Particular Solutions

- **General solution**: a solution containing arbitrary constant(s).
- **Particular solution**: a solution where the constant(s) are found from given conditions.

In this chapter, C and D denote arbitrary constants that are added, and A denotes an arbitrary constant that multiplies.

13.2.1 $\frac{dy}{dx} = f(x)$

Example 13.1	Example 13.2	Example 13.3
$\frac{dy}{dx} = 3x^2$	$\frac{dy}{dx} = e^{2x}$	$\frac{d^2y}{dx^2} = 6x$
$\vdots$	$\vdots$	$\vdots$
$y = x^3 + C$	$y = \frac{1}{2}e^{2x} + C$	$y = x^3 + Cx + D$

13.2.2 Method of Separable Variables

When a differential equation has "y" terms, we apply the method of separable variable. Divide the expression in terms of y to the LHS, and keep all x terms on the RHS.

$$f(y)\frac{dy}{dx} = g(x)$$

FLAG N28 [MINOR] — for Maximus to review: 13.2.2 Method of Separable Variables: only the form $f(y)\frac{dy}{dx}=g(x)$ is shown. The general separable form $\frac{dy}{dx}=f(x)g(y)$ (which is what Example 13.3 is) should be stated, with the rearrangement to $\frac{1}{g(y)}\frac{dy}{dx}=f(x)$.

Example 13.4

$$\frac{dy}{dx}=k(30-y)$$

$$\frac{1}{30-y}\frac{dy}{dx}=k$$

$$\vdots$$

$$\therefore y=30-Ae^{-kx}$$

Example 13.5

$$\frac{dy}{dx}=ky(1-y)$$

$$\frac{1}{y(1-y)}\frac{dy}{dx}=k$$

$$\vdots$$

$$\therefore y=\frac{1}{1+Ae^{-kx}}$$

Example 13.6

$$\frac{dy}{dx}=xy^2+x+y^2+1$$

$$=x(y^2+1)+(y^2+1)$$

$$=(x+1)(y^2+1)$$

$$\frac{1}{y^2+1}\frac{dy}{dx}=x+1$$

$$\vdots$$

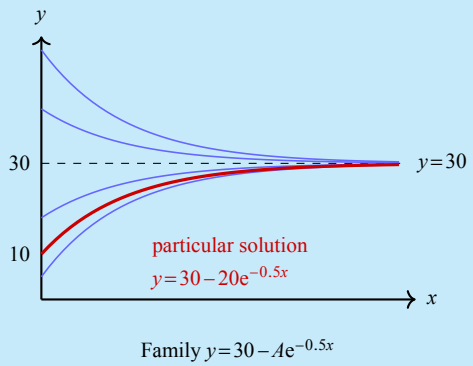
$$\therefore y=\tan\left(\frac{1}{2}x^2+x+C\right)$$

Note

If some values of x and y are given (usually initial values $x=0$, $y=y_0$), you should solve for the particular solution.

Note: Sketching graphs of solutions

- **General solution**: family of curves, i.e. many possible different curves.
- **Particular solution**: one curve.



Family $y=30-Ae^{-0.5x}$

Note

Most of the time, graphs of solutions to differential equations in H2 Math have horizontal asymptotes. We can find them by letting $x\to\infty$. Alternatively, let $\frac{dy}{dx}=0$ and solve for y if it is a separable variable differential equation.

FLAG N29 [STATEMENT] — for Maximus to review: 13.2.2 Note (asymptote shortcut): "let $\frac{dy}{dx}=0$ and solve for y if it is separable works only when $\frac{dy}{dx}$ depends on y alone (autonomous, e.g. $k(30-y)$). For a general separable $f(x)g(y)$ this also gives $f(x)=0$ (not an asymptote), and solutions need not tend to a constant. Say "if $\frac{dy}{dx}$ is a function of y only; otherwise let $x\to\infty$ in the solution.

13.3 Substitution

Given a differential equation, it may sometimes be impossible to separate the variables. In such cases, a substitution will be given to you (do not come up with your own).

Example 13.7

(DHS 2025 Prelim P1 Q1) It is given that $2xy\frac{\mathrm{d}y}{\mathrm{d}x}=x^2-y^2$ where $x>0, y<0$. Using the substitution $u=\frac{y}{x}$, show that the differential equation can be transformed to $\frac{2u}{1-3u^2}\frac{\mathrm{d}u}{\mathrm{d}x}=\frac{1}{x}$. Hence find the general solution of y in terms of x.

Step 1: Differentiate the substitution

$$u=\frac{y}{x} \Rightarrow y=ux$$
$$\frac{\mathrm{d}y}{\mathrm{d}x}=\frac{\mathrm{d}u}{\mathrm{d}x}x+u \quad \ldots(1)$$

Step 2: Substitute the substitution

$$2xy\frac{\mathrm{d}y}{\mathrm{d}x}=x^2-y^2$$
$$2x(ux)\left(x\frac{\mathrm{d}u}{\mathrm{d}x}+u\right)=x^2-(ux)^2$$
$$\vdots$$
$$\frac{2u}{1-3u^2}\frac{\mathrm{d}u}{\mathrm{d}x}=\frac{1}{x} \quad \text{(shown)}$$
$$\vdots$$
$$3u^2=1-\frac{A}{x^3}$$
$$\left(\frac{y}{x}\right)^2=\frac{1}{3}\left(1-\frac{A}{x^3}\right)$$
$$\therefore y=-\sqrt{\frac{x^2}{3}\left(1-\frac{A}{x^3}\right)}=-x\sqrt{\frac{1}{3}\left(1-\frac{A}{x^3}\right)} \quad (\because y<0)$$

VERY IMPORTANT NOTE

For substitution in differential equations, always remember to substitute the original variables (x and y) back into the equation!

SUPER ULTRA IMPORTANT NOTE

The variable in the substitution is **NOT A CONSTANT** unless otherwise stated!

Example: $y=ux^2$ (product rule: u and x^2 are both variables)

✗ $\frac{\mathrm{d}y}{\mathrm{d}x}=2ux$ ✓ $\frac{\mathrm{d}y}{\mathrm{d}x}=x^2\frac{\mathrm{d}u}{\mathrm{d}x}+u(2x)=x^2\frac{\mathrm{d}u}{\mathrm{d}x}+2ux$

14 Complex Numbers

14.1 Cartesian Form

Definition 14.1.1. Cartesian form of a complex number

Every complex number $z \in \mathbb{C}$ can be expressed as $z = x + y\mathrm{i}$, $x, y \in \mathbb{R}$.

- A complex number z is **real** if $\mathrm{Im}(z) = 0$.
- A complex number z is **purely imaginary** if $\mathrm{Re}(z) = 0$.

FLAG N30 [STATEMENT] — for Maximus to review: 14.1 Cartesian Form: "purely imaginary if $\mathrm{Re}(z) = 0$ should also require $\mathrm{Im}(z) \neq 0$; as written 0 would be both real and purely imaginary. Write "$\mathrm{Re}(z) = 0$ and $\mathrm{Im}(z) \neq 0$.

14.2 Complex Number Arithmetic

- **Addition & subtraction**: similar to surds.
- **Multiplication**: similar to surds.
- **Division**: similar to surds, remember to *realise*.
- **Square roots** (See Example 14.1)

Example 14.1

Find the values of $\sqrt{3+4\mathrm{i}}$.

$$(a+b\mathrm{i})^2 = 3+4\mathrm{i}$$
$$a^2 - b^2 + 2ab\,\mathrm{i} = 3+4\mathrm{i}$$

By comparing real & imaginary parts,

$$a^2 - b^2 = 3, \quad 2ab = 4$$
$$\vdots$$
$$a = 2,\ b = 1 \text{ or } a = -2,\ b = -1$$
$$\therefore \sqrt{3+4\mathrm{i}} = 2+\mathrm{i} \text{ or } -2-\mathrm{i}$$

FLAG N31 [MINOR] — for Maximus to review: 14.2 Complex Number Arithmetic, Example: "$\sqrt{3+4\mathrm{i}}$ … $= 2+\mathrm{i}$ or $-2-\mathrm{i}$ — the symbol $\sqrt{\ }$ is not defined for complex numbers in H2 (it is single-valued for reals). Phrase it as "find the square roots of $3+4\mathrm{i}$ / "solve $z^2 = 3+4\mathrm{i}$.

Equality of Complex Numbers

Two complex numbers $z_1 = a_1 + b_1\text{i}$ and $z_2 = a_2 + b_2\text{i}$ are equal if and only if their real parts are equal and their imaginary parts are equal.

$$z_1 = z_2 \iff a_1 = a_2 \text{ and } b_1 = b_2$$

This is basically comparing **real** and **imaginary** parts.

14.3 Simultaneous Equations

Common types:

1. Standard z, w (no conjugate, no modulus, no square)

$$2z+w = 2-4\text{i}$$
$$\text{i}z - w = 2\text{i}$$

[TJC 2025 Prelim P1 Q8(a)]

FLAG N32 [ERROR] — for Maximus to review: 14.3 Simultaneous Equations, type 1: the example is cited as TJC 2025 Prelim P1 Q8(a), but the actual TJC question is $2z+|w|=2-4\text{i}$, $\text{i}z-w=2\text{i}$ — it has a modulus $|w|$ (so it belongs under type 4), not $2z+w$. Either cite it as "adapted from or use the original under the modulus type. (The equations shown here, $2z+w$, are not what the source asks.)

2. With z^* and w (or vice versa, or both)

$$z+2z^*+w = 6+\text{i} \quad \text{and} \quad \text{i}z - w = 3$$

[MI 2025 Prelim P2 Q3(a)]

If both z^* and w^* are in the equation, it is still solvable. Consider this example: $z = 2+\text{i}w \Rightarrow z^* = 2-\text{i}w^*$

3. With z^2 and w

$$v^2 - \text{i}w + 2 = 0$$
$$v = w + \text{i} + 3$$

[NYJC 2024 Prelim P1 Q1]

The quadratic formula is useful here; avoid letting $z = a+b\text{i}$ and expanding, though it works.

FLAG N33 [ERROR] — for Maximus to review: 14.3 Simultaneous Equations, type 3: the heading says "With z^2 and w but the displayed equations are in v ($v^2 - \text{i}w + 2 = 0$, $v = w + \text{i} + 3$), and the grey note then talks about $z = a+b\text{i}$. Use the same letter in the heading, the equations and the note.

4. With z and $|w|$

$$z + w^* + 5\text{i} = 10 \quad \text{and} \quad |w|^2 = z + 18 + \text{i}$$

[CJC 2017 Prelim P2 Q4(a)]

For $w=a+b\text{i}$, $|w|=\sqrt{a^2+b^2}$.

Note

Please check your answers using your GC for **BOTH** equations! The answers are usually quite nice and not ugly numbers.

14.4 Polynomials

Complex Conjugate Root Theorem

If all the coefficients of a polynomial equation $P(z)=0$ are real, and $z=\alpha$ is a root, then its conjugate $z=\alpha^*$ is also a root.

Example 14.2

[ACJC/2014/I/4]

(i) Given that $1+\mathrm{i}$ is a root of the equation $2w^3+aw^2+bw-2=0$, find the values of the real numbers a and b. [4]

(ii) For these values of a and b, solve the equation in part **(ii)**, without the use of a calculator. [3]

FLAG N34 [ERROR] — for Maximus to review: 14.4 Polynomials, ACJC/2014/I/4: part (ii) says "solve the equation in part **(ii)** — it refers to itself. It should refer to the equation in part (i). Sourced question text: fix against the original paper.

Method 1: Apply CCRT

$w=1+\mathrm{i}$ is a root

$\therefore w=1-\mathrm{i}$ is a root

$$(w-(1+\mathrm{i}))\,(w-(1-\mathrm{i}))=(w-1)^2-\mathrm{i}^2$$
$$=w^2-2w+2$$
$$2w^3+aw^2+bw-2=(w^2-2w+2)\,(Aw+B)$$

w^3: $A=2$

w^0: $2B=-2$

$B=-1$

$$Aw+B=2w-1=0$$
$$\therefore w=\frac{1}{2}$$

w^2: $a=B-2A=-5$

w^1: $b=2A-2B=6$

$\therefore a=-5,\ b=6$

$\therefore w=1+\mathrm{i},\ 1-\mathrm{i},\ \frac{1}{2}$

Method 2: Substitute the root

$$2(1+\mathrm{i})^3+a(1+\mathrm{i})^2+b(1+\mathrm{i})-2=0$$
$$(b-6)+(2a+b+4)\mathrm{i}=0$$

By comparing real & imaginary parts,

$$b-6=0,\quad 2a+b+4=0$$
$$\therefore a=-5,\ b=6$$

$w=1+\mathrm{i}$ is a root

$\therefore w=1-\mathrm{i}$ is a root

$$(w-(1+\mathrm{i}))\,(w-(1-\mathrm{i}))=w^2-2w+2$$
$$2w^3-5w^2+6w-2=(w^2-2w+2)\,(Aw+B)$$

(compare coefficients or long divide) ⋮

$A=2,\ B=-1$

$\therefore w=1+\mathrm{i},\ 1-\mathrm{i},\ \frac{1}{2}$

Note

If you see a really weird question you have never seen before, you can try letting the complex number $=a+b\mathrm{i}$ as a last resort and see where you get with it (unless it becomes unreasonably tedious).

Example 14.3

[RI Promo/9758/2018/Q11(a)(ii)] It is given that one of the roots of the equation $z^3-(1+4\mathrm{i})z^2-(3-9\mathrm{i})z+(14-2\mathrm{i})=0$ is purely imaginary. Find all the roots of the equation, giving your answers in Cartesian form $a+\mathrm{i}b$. [6]

The coefficients are not all real, so the Complex Conjugate Root Theorem does not apply.

$$\text{Let } z=k\mathrm{i},\ k\in\mathbb{R}$$

$$-k^3\mathrm{i}+(1+4\mathrm{i})k^2-(3-9\mathrm{i})k\mathrm{i}+14-2\mathrm{i}=0$$

$$(k^2-9k+14)+(-k^3+4k^2-3k-2)\,\mathrm{i}=0$$

By comparing real & imaginary parts,

$$k^2-9k+14=0,\quad -k^3+4k^2-3k-2=0$$

$$(k-2)(k-7)=0$$

$$k=2 \text{ or } 7$$

$$k=2:\quad -8+16-6-2=0$$

$$k=7:\quad -343+196-21-2\neq 0$$

$$\therefore k=2,\quad z=2\mathrm{i}$$

$$z^3-(1+4\mathrm{i})z^2-(3-9\mathrm{i})z+(14-2\mathrm{i})=(z-2\mathrm{i})\,(Az^2+Bz+C)$$

Here, $A, B, C\in\mathbb{C}$, so we cannot compare real and imaginary parts for an equation involving A, B or C.

$$z^3:\quad A=1$$

$$z^2:\quad B-2\mathrm{i}A=-1-4\mathrm{i}$$

$$B=-1-2\mathrm{i}$$

$$z^0:\quad -2\mathrm{i}C=14-2\mathrm{i}$$

$$C=1+7\mathrm{i}$$

$$z^2-(1+2\mathrm{i})z+(1+7\mathrm{i})=0$$

$$z=\frac{(1+2\mathrm{i})\pm\sqrt{(1+2\mathrm{i})^2-4(1+7\mathrm{i})}}{2}$$

$$=\frac{(1+2\mathrm{i})\pm\sqrt{-7-24\mathrm{i}}}{2}\quad \text{(use GC here!)}$$

$$=\frac{(1+2\mathrm{i})\pm(3-4\mathrm{i})}{2}$$

$$\therefore z=2\mathrm{i},\ 2-\mathrm{i},\ -1+3\mathrm{i}$$

This question does not say "without using a calculator", so use your GC to evaluate anything you can, instead of expanding things out manually.

14.5 Argand Diagram

14.5.1 Modulus

Definition 14.5.1. Modulus of a Complex Number

The **Modulus** of complex number, or absolute value, is defined as:

$$|z| = |a+b\mathrm{i}| = \sqrt{a^2+b^2}.$$

It is represented by the distance from the origin, to the coordinate represented by the complex number in an Argand diagram.

Properties of Modulus

Let $z=a+b\mathrm{i}$ and $w=x+y\mathrm{i}$, where $a, b, x, y \in \mathbb{R}$. Then we have the following useful properties.

- $|z|^2 = a^2+b^2$. (Obviously)
- $|z^*| = |z|$.
- $|zw| = |z||w|$.
- $\left|\frac{z}{w}\right| = \frac{|z|}{|w|}$.
- $|z^n| = |z|^n$.

You are not required to memorise the last 3 formulas in the 'A' Level syllabus, but may be asked to show them and use it in a question.

FLAG N35 [STATEMENT] — for Maximus to review: 14.5.1 Modulus: "not required to memorise the last 3 formulas" — the 9758 syllabus (4.2) includes multiplication and division of complex numbers in polar form, so $|zw|=|z||w|$ and $\left|\frac{z}{w}\right|=\frac{|z|}{|w|}$ (and the matching argument rules) are examinable, not just "may be asked to show". Only $|z^n|=|z|^n$ is not listed explicitly. Reword.

14.5.2 Argument

Definition 14.5.2. Argument of a Complex Number

An **argument** of the complex number $z=x+\mathrm{i}y$, denoted $\arg(z)$, is defined geometrically, as the 2D polar angle θ from the positive real axis to the vector representing z.

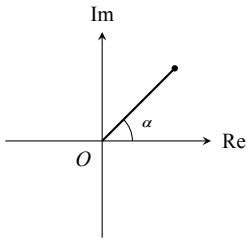
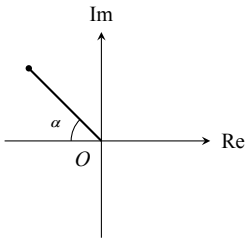
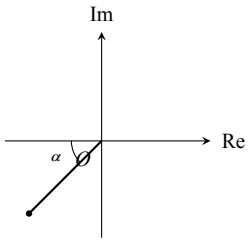
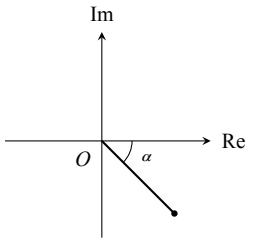
Note

The argument, θ, of all complex numbers must lie within the range $-\pi < \theta \leq \pi$.

Given a complex number, to find its argument,

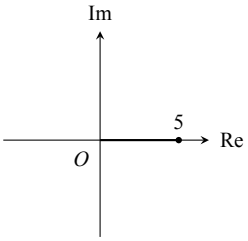
Step 1: find basic angle $\alpha = \tan^{-1}\left|\frac{b}{a}\right|$.

Step 2: draw a small Argand diagram to figure out which quadrant it is in based on its coordinate. (e.g. $-1+\sqrt{3}\,\mathrm{i} \iff (-1, \sqrt{3})$ is in the 2nd quadrant)

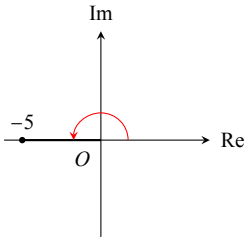
1st Quadrant	2nd Quadrant	3rd Quadrant	4th Quadrant
			
$\arg(z)=\alpha$	$\arg(z)=\pi-\alpha$	$\arg(z)=-(\pi-\alpha)$	$\arg(z)=-\alpha$

Caution

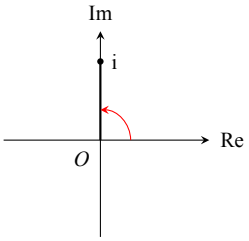
For purely real or imaginary numbers, it is recommended to find the argument of them via considering their position on the Argand diagram.



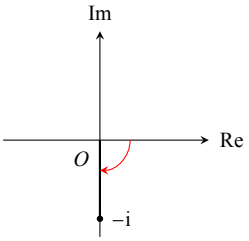
$\arg(5)=0$



$\arg(-5)=\pi$



$\arg(\mathrm{i})=\frac{\pi}{2}$



$\arg(-\mathrm{i})=-\frac{\pi}{2}$

Properties of Argument

Let $\arg(z)=\theta_z$ and $\arg(w)=\theta_w$. Then we have the following useful properties.

- $\arg(z^*)=-\arg(z)$.

- $\arg(zw)=\arg(z)+\arg(w)$ $=\theta_z+\theta_w$.
- $\arg\left(\frac{z}{w}\right)=\arg(z)-\arg(w)$ $=\theta_z-\theta_w$.
- $\arg(z^n)=n\cdot\arg(z)=n\theta_z$.

You are not required to memorise the last 3 formulas in the 'A' Level syllabus, but they may be given to you and you have to use them to answer the rest of the question.

FLAG N36 [STATEMENT] — for Maximus to review: 14.5.2 Properties of Argument: $\arg(zw)=\arg z+\arg w$, $\arg\frac{z}{w}=\arg z-\arg w$ and $\arg(z^n)=n\arg z$ hold only modulo 2π — the sum/difference/multiple can fall outside $(-\pi,\pi]$ and must then be adjusted by $\pm2\pi$. Also $\arg(z^*)=-\arg(z)$ fails when z is a negative real: $\arg(-5)=\pi$ but $-\arg(-5)=-\pi$ is outside the range (use π). State "(mod 2π) and the exception.

14.5.3 Finding Cartesian Form Given Modulus and Argument

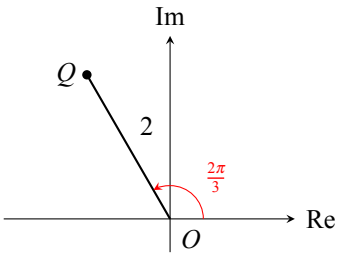
Example 14.4

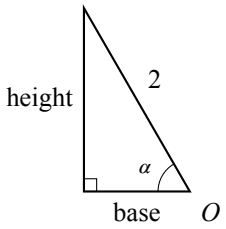
[ASRJC 2025 Prelims P2 Q6] The point Q is represented by the complex number q, where

$$\arg(q)=\frac{2}{3}\pi \quad \text{and} \quad |q|=2.$$

(a) Sketch the point Q on an Argand diagram. [2]

(b) Use the Argand diagram to deduce $\text{Re}(q)$ and $\text{Im}(q)$, giving your answers in exact form. [2]





$$\alpha=\pi-\frac{2\pi}{3}=\frac{\pi}{3}$$
$$\text{base}=2\cos\frac{\pi}{3}=1$$
$$\text{height}=2\sin\frac{\pi}{3}=\sqrt{3}$$
$$\therefore \text{Re}(q)=-1, \quad \text{Im}(q)=\sqrt{3}$$

Q is in the 2nd quadrant, so the real part is negative.

Common Mistake

Let $q=a+b\text{i}$,

$$\sqrt{a^2+b^2}=2, \quad \tan^{-1}\frac{b}{a}=\frac{2\pi}{3} \quad \text{✗✗}$$

This is very tedious to solve, and the argument equation is also wrong; we can't simply take $\tan^{-1}\frac{b}{a}$ to find the argument for any complex number. We need to first find the basic angle then consider which quadrant it is in.

14.5.4 Addition and Scalar Multiplication in an Argand Diagram

Complex numbers in an Argand diagram behave like 2 dimensional vectors. When adding two complex numbers (vectors) together, the result follows the parallelogram law of vector addition. Similarly, multiplying a complex number (vector) by a scalar constant scales its length, and if the scalar is negative, it reflects the complex number (vector) about the origin.

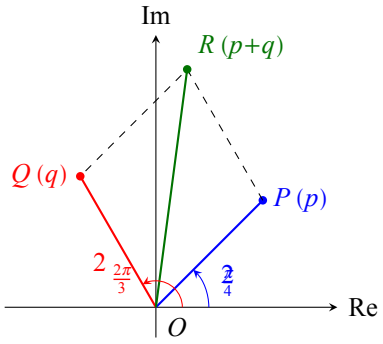
Example 14.5

[Modified from ASRJC 2025 Prelims P2 Q6] The points P, Q and R correspond to the complex numbers p, q and $p+q$ respectively. Given $p=\sqrt{2}+\sqrt{2}\,\mathrm{i}$, $q=-1+\sqrt{3}\,\mathrm{i}$.

(a) Sketch P, Q and R on a single Argand diagram.

(b) By considering $\arg(p+q)$ or otherwise, show that $\tan\left(\frac{11}{24}\pi\right)=\sqrt{6}+\sqrt{3}+\sqrt{2}+2$. [3]

(a)



Note: $p+q=\left(\sqrt{2}-1\right)+\left(\sqrt{2}+\sqrt{3}\right)\mathrm{i}$

Here, we use vector addition to determine the position of R, rather than its coordinate $\left(\sqrt{2}-1,\ \sqrt{2}+\sqrt{3}\right)$.

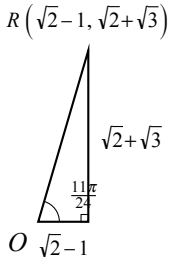
(b)

$|p|=|q|=2$, so $OPRQ$ is a rhombus and OR bisects angle POQ.

$$\because |p|=|q|=2$$

$$\arg(p+q)=\frac{1}{2}\left(\frac{\pi}{4}+\frac{2\pi}{3}\right)=\frac{11\pi}{24}$$

By TOA CAH SOH,



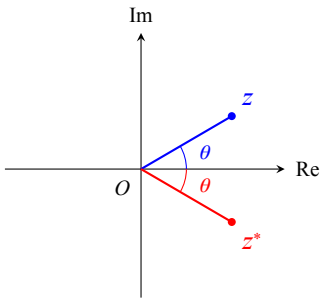
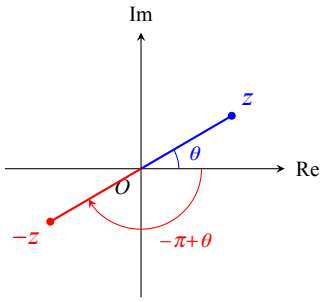
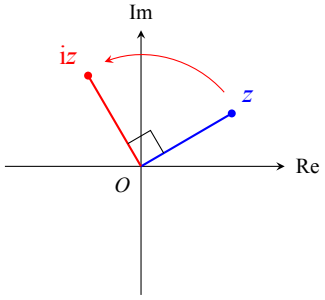
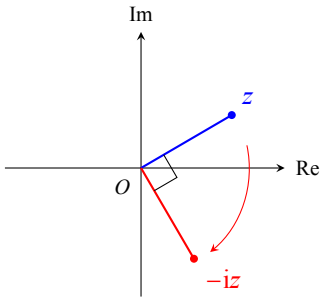
$$\tan\left(\frac{11\pi}{24}\right)=\frac{\sqrt{2}+\sqrt{3}}{\sqrt{2}-1}$$

$$=\frac{\left(\sqrt{2}+\sqrt{3}\right)\left(\sqrt{2}+1\right)}{\left(\sqrt{2}-1\right)\left(\sqrt{2}+1\right)}$$

$$=2+\sqrt{2}+\sqrt{6}+\sqrt{3}$$

$$\therefore \tan\left(\frac{11\pi}{24}\right)=\sqrt{6}+\sqrt{3}+\sqrt{2}+2 \quad \text{(shown)}$$

14.5.5 Transformations on the Argand Diagram

Transformation	Diagram	Description
$z \to z^*$		Reflect about real axis
$z \to -z$		Reflect about origin
$z \to \mathrm{i}z$		Rotate anticlockwise by 90°
$z \to -\mathrm{i}z$		Rotate clockwise by 90°

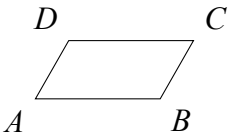
FLAG N37 [MINOR] — for Maximus to review: 14.5.5 Transformations on the Argand Diagram: only $z \to z^*$, $-z$, $\mathrm{i}z$, $-\mathrm{i}z$ are shown. The syllabus also expects the geometrical effect of addition (translation $z+c$), multiplication by a real constant (scaling kz) and by $\cos\varphi + \mathrm{i}\sin\varphi$ (rotation by φ), and division. Add these rows.

FLAG N38 [OMISSION] — for Maximus to review: Chapter 14 Complex Numbers is incomplete (the file header says "more to come): polar/exponential form $r(\cos\theta + \mathrm{i}\sin\theta)$, $re^{\mathrm{i}\theta}$, multiplication and division in polar form (syllabus 4.2) and simple loci are not yet in the notes.

15 Vectors

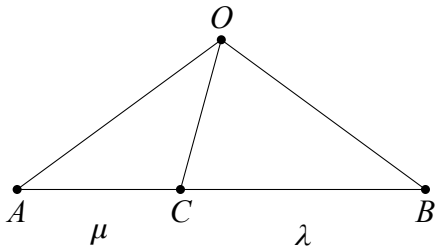
Vectors I

15.1 Basic Formulas

	Formula	Note
Magnitude	$\lvert\mathbf{a}\rvert = \sqrt{x^2+y^2+z^2}$	
Unit vector	$\hat{\mathbf{a}} = \dfrac{\mathbf{a}}{\lvert\mathbf{a}\rvert}$	$\lvert\hat{\mathbf{a}}\rvert = 1$
Parallel vectors	$\mathbf{a} = k\mathbf{b}$ $\quad \mathbf{a}\times\mathbf{b} = \mathbf{0}$	k scalar constant
Collinear A, B, C	$\overrightarrow{AB} \parallel \overrightarrow{BC}$ (or any two of $\overrightarrow{AB}, \overrightarrow{BC}, \overrightarrow{AC}$)	Try drawing the diagram and take any two vectors, to show that they are parallel.
Parallelogram $ABCD$	$\overrightarrow{AB} = \overrightarrow{DC}$, $\quad \overrightarrow{AD} = \overrightarrow{BC}$	

The Ratio Theorem

Let the points A, B and C have position vectors $\mathbf{a}$, $\mathbf{b}$ and $\mathbf{c}$ respectively, and suppose that $\lambda AC = \mu CB$. Then the position vector of C is given by the formula

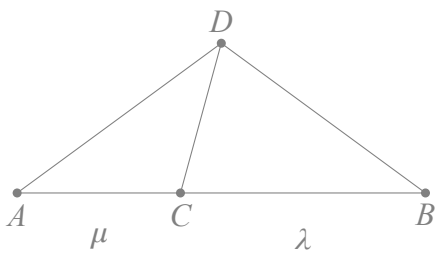


Ratio Theorem

$$\mathbf{c} = \frac{\lambda\mathbf{a} + \mu\mathbf{b}}{\lambda + \mu}$$

$AC : CB = \mu : \lambda$, so C divides AB with μ beside A and λ beside B.

The same theorem, starting from a point D instead of O:

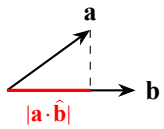
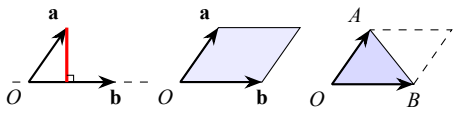


Ratio Theorem

$$\overrightarrow{DC} = \frac{\lambda\overrightarrow{DA} + \mu\overrightarrow{DB}}{\lambda + \mu}$$

15.2 Scalar Product and Vector Product

$$\mathbf{a}=a_1\mathbf{i}+a_2\mathbf{j}+a_3\mathbf{k}, \qquad \mathbf{b}=b_1\mathbf{i}+b_2\mathbf{j}+b_3\mathbf{k}, \qquad \theta=\text{non-reflex } \angle AOB$$

	Scalar Product (Dot)	Vector Product (Cross)
Definition	$\mathbf{a}\cdot\mathbf{b}=a_1b_1+a_2b_2+a_3b_3$	$\mathbf{a}\times\mathbf{b}=\begin{pmatrix} a_2b_3-a_3b_2 \\ -(a_1b_3-a_3b_1) \\ a_1b_2-a_2b_1 \end{pmatrix}$
Result	scalar	vector, $\perp$ to both $\mathbf{a}$ and $\mathbf{b}$
Formula	$\mathbf{a}\cdot\mathbf{b}=\lvert\mathbf{a}\rvert\lvert\mathbf{b}\rvert\cos\theta$	$\lvert\mathbf{a}\times\mathbf{b}\rvert=\lvert\mathbf{a}\rvert\lvert\mathbf{b}\rvert\sin\theta$
Angle	$\cos\theta=\dfrac{\mathbf{a}\cdot\mathbf{b}}{\lvert\mathbf{a}\rvert\lvert\mathbf{b}\rvert}$ $\mathbf{a}\cdot\mathbf{b}>0$ acute, <0 obtuse, $=0$ right angle	$\sin\theta=\dfrac{\lvert\mathbf{a}\times\mathbf{b}\rvert}{\lvert\mathbf{a}\rvert\lvert\mathbf{b}\rvert}$ this method only finds the basic angle possible, but usually undesirable
Properties	$\mathbf{a}\cdot\mathbf{b}=\mathbf{b}\cdot\mathbf{a}$ $\mathbf{a}\cdot(\mathbf{b}+\mathbf{c})=\mathbf{a}\cdot\mathbf{b}+\mathbf{a}\cdot\mathbf{c}$ $\lambda(\mathbf{a}\cdot\mathbf{b})=(\lambda\mathbf{a})\cdot\mathbf{b}=\mathbf{a}\cdot(\lambda\mathbf{b})$ $\mathbf{a}\cdot\mathbf{a}=\lvert\mathbf{a}\rvert^2$	$\mathbf{a}\times\mathbf{b}=-\mathbf{b}\times\mathbf{a}$ $\mathbf{a}\times(\mathbf{b}+\mathbf{c})=\mathbf{a}\times\mathbf{b}+\mathbf{a}\times\mathbf{c}$ $\lambda(\mathbf{a}\times\mathbf{b})=(\lambda\mathbf{a})\times\mathbf{b}=\mathbf{a}\times(\lambda\mathbf{b})$ $\mathbf{a}\times\mathbf{a}=\mathbf{0}$
Test	$\mathbf{a}\cdot\mathbf{b}=0 \iff \mathbf{a}\perp\mathbf{b}$	$\mathbf{a}\times\mathbf{b}=\mathbf{0} \iff \mathbf{a}\parallel\mathbf{b}$
	(nonzero $\mathbf{a}$, $\mathbf{b}$)	
Applications	Length of projection $=\lvert\mathbf{a}\cdot\hat{\mathbf{b}}\rvert$ Projection vector $=(\mathbf{a}\cdot\hat{\mathbf{b}})\hat{\mathbf{b}}$	Perpendicular distance $=\lvert\mathbf{a}\times\hat{\mathbf{b}}\rvert$ Area of $\triangle OAB=\frac{1}{2}\lvert\mathbf{a}\times\mathbf{b}\rvert$ Area of parallelogram $=\lvert\mathbf{a}\times\mathbf{b}\rvert$
Geometrical meaning	 $\lvert\mathbf{a}\cdot\hat{\mathbf{b}}\rvert$ = length of projection of $\mathbf{a}$ onto $\mathbf{b}$	 perpendicular distance from A to line along $\mathbf{b}$ $\lvert\mathbf{a}\times\hat{\mathbf{b}}\rvert$ area of parallelogram, sides $\mathbf{a}$, $\mathbf{b}$ $\lvert\mathbf{a}\times\mathbf{b}\rvert$ 2× area of $\triangle OAB$ $\lvert\mathbf{a}\times\mathbf{b}\rvert$
Note	$\mathbf{a}\cdot\mathbf{b}\cdot\mathbf{c}$ meaningless; no cancellation law	no cancellation law $(\mathbf{a}\times\mathbf{b})\times\mathbf{c}$ is a vector, but triple cross products are usually not tested

Note

Projection of $\mathbf{a}$ onto $\mathbf{b}$ $\neq$ projection of $\mathbf{b}$ onto $\mathbf{a}$. Perpendicular distance from A to OB $\neq$ from B to OA. Triangles without O as a vertex: $\frac{1}{2}\lvert\overrightarrow{AB}\times\overrightarrow{AC}\rvert$.

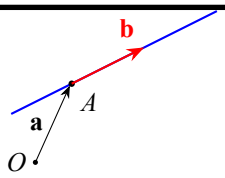
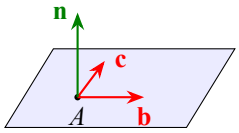
Useful identity

For any vectors $\mathbf{a}$, $\mathbf{b}$: $(\mathbf{a}\times\mathbf{b})\cdot\mathbf{a}=0$ and $(\mathbf{a}\times\mathbf{b})\cdot\mathbf{b}=0$

$\because \mathbf{a}\times\mathbf{b}$ is perpendicular to $\mathbf{a}$ and $\mathbf{b}$.

Vectors II

15.3 Equations of Lines and Planes

	Vector	Cartesian	What you need
Line	 $\mathbf{r}=\mathbf{a}+\lambda\mathbf{b},\ \lambda\in\mathbb{R}$ $(\mathbf{r}-\mathbf{a})\times\mathbf{b}=\mathbf{0}$ (same line: $\mathbf{r}-\mathbf{a}$ // $\mathbf{b}$)	$\frac{x-a_x}{b_x}=\frac{y-a_y}{b_y}=\frac{z-a_z}{b_z}$ $\mathbf{a}=\begin{pmatrix}a_x\\a_y\\a_z\end{pmatrix}$, $\mathbf{b}=\begin{pmatrix}b_x\\b_y\\b_z\end{pmatrix}$ if $b_x=0$: $x=a_x$ separately all coefficients of x, y and z in the numerators must be 1	**Need:** 1 point + 1 direction or 2 points A, B: $\mathbf{b}=\overrightarrow{AB}$
Plane	 $\mathbf{r}=\mathbf{a}+\lambda\mathbf{b}+\mu\mathbf{c}$ $\mathbf{r}\cdot\mathbf{n}=d$ (scalar product form)	$n_xx+n_yy+n_zz=d$ $\mathbf{n}=\begin{pmatrix}n_x\\n_y\\n_z\end{pmatrix}$	**Need:** 1 point + 2 non-parallel directions // plane or 1 point + normal $\mathbf{n}$ $\mathbf{n}=\mathbf{b}\times\mathbf{c}$, $d=\mathbf{a}\cdot\mathbf{n}$ 3 points: $\mathbf{b}=\overrightarrow{AB}$, $\mathbf{c}=\overrightarrow{AC}$; contains a line: use its direction

$\mathbf{a}$: position vector of any point on the line/plane. $\mathbf{b}$, $\mathbf{c}$: any two linearly independent directions parallel to the plane. The equation of any line is not unique.

Note

To check if a given point C lies on
- a line: $\overrightarrow{OC}=\mathbf{a}+\lambda\mathbf{b}$ (one λ satisfies all 3 components)
- a plane: $\overrightarrow{OC}\cdot\mathbf{n}=d$

Example 15.1

Line through $(1,2,3)$, $(3,2,2)$

$$\mathbf{b}=\begin{pmatrix}2\\0\\-1\end{pmatrix}$$

$$\mathbf{r}=\begin{pmatrix}1\\2\\3\end{pmatrix}+\lambda\begin{pmatrix}2\\0\\-1\end{pmatrix},\quad \lambda\in\mathbb{R}$$

$$\frac{x-1}{2}=\frac{z-3}{-1},\quad y=2$$

Example 15.2

Plane through $B(0,1,1)$, $C(1,1,3)$, $D(2,0,2)$

$$\mathbf{n}=\overrightarrow{BC}\times\overrightarrow{BD}=\begin{pmatrix}1\\0\\2\end{pmatrix}\times\begin{pmatrix}2\\-1\\1\end{pmatrix}=\begin{pmatrix}2\\3\\-1\end{pmatrix}$$

$$d=\begin{pmatrix}0\\1\\1\end{pmatrix}\cdot\begin{pmatrix}2\\3\\-1\end{pmatrix}=2$$

$$2x+3y-z=2$$

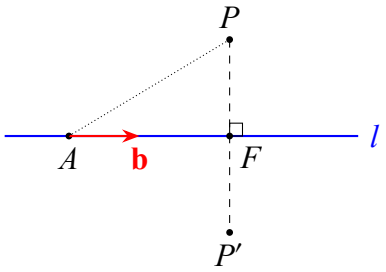
15.4 Interactions Map

	Point	Line	Plane
Point	distance $\lvert\overrightarrow{AB}\rvert$		
Line	• on line? §2.1 • foot F §2.3 • shortest distance §2.3 • reflection §2.3	• // / intersect / skew §2.4 • point of intersection §2.4 • angle §2.6 • reflect line in line §2.7	
Plane	• on plane? §2.1 • foot F §2.3 • shortest distance §2.3 • reflection §2.3	• // / lies in / intersect §2.4 • point of intersection §2.4 • angle §2.6 • reflect line in plane §2.7	• // / same / intersect §2.4 • line of intersection §2.4 • angle §2.6 • distance (//) §2.8

Always draw the diagram first.

15.5 Foot / Distance / Reflection

Point P – Line $\mathbf{r}=\mathbf{a}+\lambda\mathbf{b}$

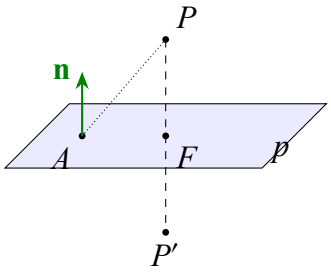


1. $\overrightarrow{OF}=\mathbf{a}+\lambda\mathbf{b}$ for some $\lambda\in\mathbb{R}$
2. $\overrightarrow{PF}\cdot\mathbf{b}=0$ $\Rightarrow$ solve for λ
3. $\overrightarrow{OF}=\dfrac{\overrightarrow{OP}+\overrightarrow{OP'}}{2}$ (midpoint / ratio theorem)

Distance

$$PF=\lvert\overrightarrow{AP}\times\hat{\mathbf{b}}\rvert$$

Point P – Plane $\mathbf{r}\cdot\mathbf{n}=d$



1. $l_{PF}:\ \mathbf{r}=\overrightarrow{OP}+\lambda\mathbf{n},\ \lambda\in\mathbb{R}$
2. $(\overrightarrow{OP}+\lambda\mathbf{n})\cdot\mathbf{n}=d$ (intersection between line & plane) $\Rightarrow$ solve for λ
3. $\overrightarrow{OF}=\dfrac{\overrightarrow{OP}+\overrightarrow{OP'}}{2}$ (midpoint / ratio theorem)

Distance

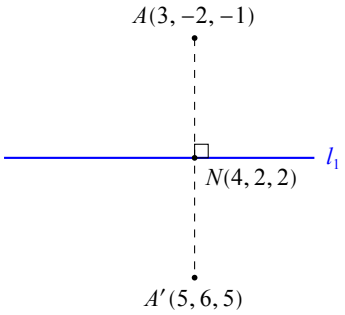
$$PF=\lvert\overrightarrow{AP}\cdot\hat{\mathbf{n}}\rvert$$

Reflection (and foot of perpendicular) of point to line

Example 15.3

Find the foot of the perpendicular N from $A(3,-2,-1)$ to the line l_1, and the reflection A' of A in l_1, where

$$l_1: \mathbf{r}=\begin{pmatrix}1\\2\\3\end{pmatrix}+\lambda\begin{pmatrix}3\\0\\-1\end{pmatrix}, \quad \lambda\in\mathbb{R}$$



$$\overrightarrow{AN}=\begin{pmatrix}-2+3\lambda\\4\\4-\lambda\end{pmatrix}$$

$$\overrightarrow{AN}\cdot\begin{pmatrix}3\\0\\-1\end{pmatrix}=0$$

$$10\lambda-10=0 \Rightarrow \lambda=1$$

$$\therefore N=(4,2,2)$$

$$\overrightarrow{ON}=\frac{\overrightarrow{OA}+\overrightarrow{OA'}}{2} \quad \text{(midpoint/ratio theorem)}$$

$$\Rightarrow \overrightarrow{OA'}=2\overrightarrow{ON}-\overrightarrow{OA}$$

$$\overrightarrow{OA'}=2\begin{pmatrix}4\\2\\2\end{pmatrix}-\begin{pmatrix}3\\-2\\-1\end{pmatrix}$$

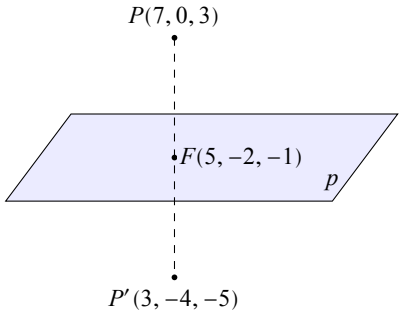
$$\therefore A'=(5,6,5)$$

Reflection (and foot of perpendicular) of point to plane

Example 15.4

Find the foot of the perpendicular F from $P(7,0,3)$ to the plane p, and the reflection P' of P in p, where

$$p: \mathbf{r}\cdot\begin{pmatrix}1\\1\\2\end{pmatrix}=1$$



$$l_{PF}: \mathbf{r}=\begin{pmatrix}7\\0\\3\end{pmatrix}+\lambda\begin{pmatrix}1\\1\\2\end{pmatrix}, \quad \lambda\in\mathbb{R}$$

$$\left(\begin{pmatrix}7\\0\\3\end{pmatrix}+\lambda\begin{pmatrix}1\\1\\2\end{pmatrix}\right)\cdot\begin{pmatrix}1\\1\\2\end{pmatrix}=1$$

$$13+6\lambda=1 \Rightarrow \lambda=-2$$

$$\therefore F=(5,-2,-1)$$

$$\overrightarrow{OF}=\frac{\overrightarrow{OP}+\overrightarrow{OP'}}{2} \quad \text{(midpoint/ratio theorem)}$$

$$\Rightarrow \overrightarrow{OP'}=2\overrightarrow{OF}-\overrightarrow{OP}$$

$$\overrightarrow{OP'}=2\begin{pmatrix}5\\-2\\-1\end{pmatrix}-\begin{pmatrix}7\\0\\3\end{pmatrix}$$

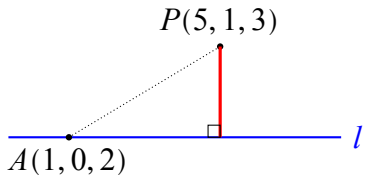
$$\therefore P'=(3,-4,-5)$$

Shortest distance from point to line

Example 15.5

Find the shortest distance from $P(5,1,3)$ to the line

$$l: \mathbf{r}=\begin{pmatrix}1\\0\\2\end{pmatrix}+\lambda\begin{pmatrix}2\\-1\\2\end{pmatrix}, \quad \lambda\in\mathbb{R}$$



$$\text{shortest distance}=|\overrightarrow{AP}\times\hat{\mathbf{b}}|$$

$$=\left|\begin{pmatrix}4\\1\\1\end{pmatrix}\times\frac{1}{3}\begin{pmatrix}2\\-1\\2\end{pmatrix}\right|$$

$$=\left|\frac{1}{3}\begin{pmatrix}3\\-6\\-6\end{pmatrix}\right|=\left|\begin{pmatrix}1\\-2\\-2\end{pmatrix}\right|$$

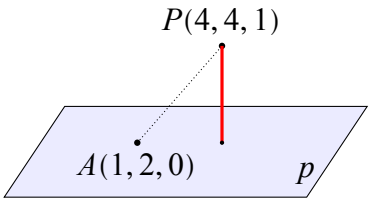
$$=3$$

Shortest distance from point to plane

Example 15.6

Find the shortest distance from $P(4,4,1)$ to the plane

$$p: \mathbf{r}\cdot\begin{pmatrix}2\\1\\-2\end{pmatrix}=4$$



$$\text{shortest distance}=|\overrightarrow{AP}\cdot\hat{\mathbf{n}}|$$

$$=\left|\begin{pmatrix}3\\2\\1\end{pmatrix}\cdot\frac{1}{3}\begin{pmatrix}2\\1\\-2\end{pmatrix}\right|$$

$$=\left|\frac{1}{3}(6+2-2)\right|$$

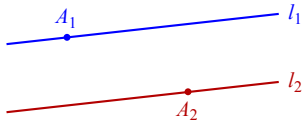
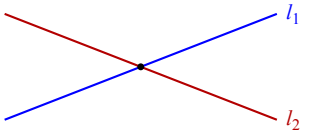
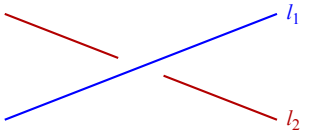
$$=2$$

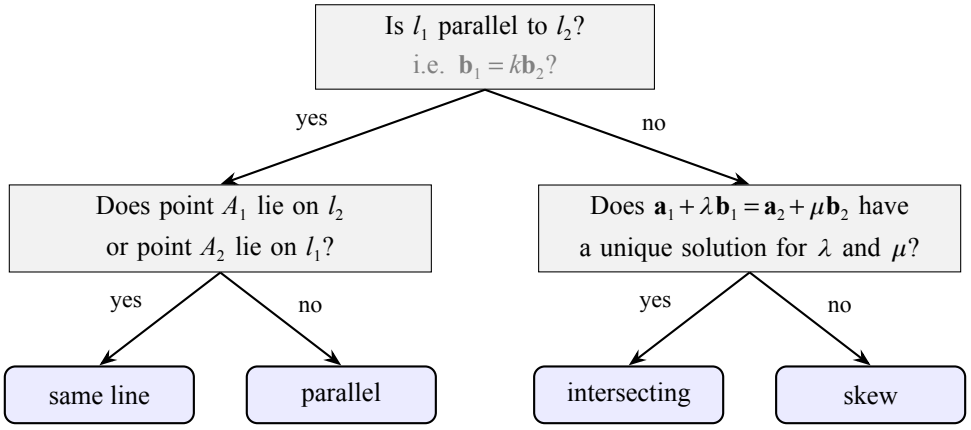
Note

We can also find the coordinates of the foot of the perpendicular N first, then take $|\overrightarrow{PN}|$, but this is less efficient.

15.6 Relationships and Intersections

Line – Line $l_1 : \mathbf{r} = \mathbf{a}_1 + \lambda \mathbf{b}_1,\ l_2 : \mathbf{r} = \mathbf{a}_2 + \mu \mathbf{b}_2$

	Parallel	Intersecting	Skew
Diagram	 $l_1 : \mathbf{r} = \mathbf{a}_1 + \lambda \mathbf{b}_1,\ \lambda \in \mathbb{R}$ $l_2 : \mathbf{r} = \mathbf{a}_2 + \mu \mathbf{b}_2,\ \mu \in \mathbb{R}$	 $l_1 : \mathbf{r} = \mathbf{a}_1 + \lambda \mathbf{b}_1,\ \lambda \in \mathbb{R}$ $l_2 : \mathbf{r} = \mathbf{a}_2 + \mu \mathbf{b}_2,\ \mu \in \mathbb{R}$	 $l_1 : \mathbf{r} = \mathbf{a}_1 + \lambda \mathbf{b}_1,\ \lambda \in \mathbb{R}$ $l_2 : \mathbf{r} = \mathbf{a}_2 + \mu \mathbf{b}_2,\ \mu \in \mathbb{R}$
Condition	$\mathbf{b}_1 \parallel \mathbf{b}_2$ and $A_1 \notin l_2, A_2 \notin l_1$	$\mathbf{b}_1 \nparallel \mathbf{b}_2$, unique solution for λ and μ	$\mathbf{b}_1 \nparallel \mathbf{b}_2$, no solution for λ and μ



Note

Definition of **skew lines**:

- non parallel
- non intersecting

Thus, to prove that two lines are skew, we must first state that $\mathbf{b}_1 \neq k\mathbf{b}_2$, so l_1 and l_2 are not parallel, before proceeding with simultaneous equations to show they have no solution.

Example 15.7

Determine whether l_1 and l_2 are intersecting or skew lines.

$$l_1: \mathbf{r} = \begin{pmatrix} 1 \\ 0 \\ 2 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix}, \quad \lambda \in \mathbb{R}$$

$$l_2: \mathbf{r} = \begin{pmatrix} 2 \\ 12 \\ 5 \end{pmatrix} + \mu \begin{pmatrix} -1 \\ 3 \\ 0 \end{pmatrix}, \quad \mu \in \mathbb{R}$$

$$1 + \lambda = 2 - \mu \qquad \text{—(1)}$$

$$2\lambda = 12 + 3\mu \qquad \text{—(2)}$$

$$2 + \lambda = 5 \qquad \text{—(3)}$$

From (1) and (3), $\lambda = 3, \quad \mu = -2$

Sub into (2), $\text{LHS} = 6, \quad \text{RHS} = 12 + 3(-2) = 6$

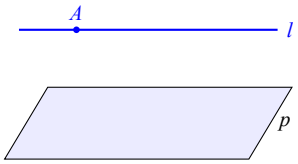
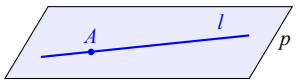
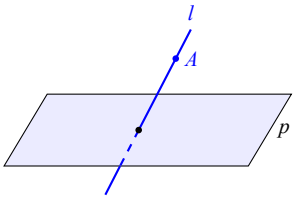
$\therefore \text{LHS} = \text{RHS}$

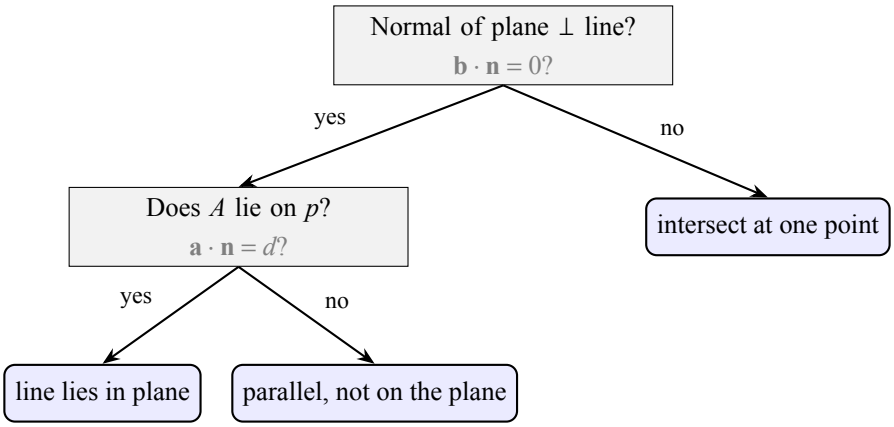
Sub $\lambda = 3$ into $l_1: \mathbf{r} = \begin{pmatrix} 1 \\ 0 \\ 2 \end{pmatrix} + 3 \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix} = \begin{pmatrix} 4 \\ 6 \\ 5 \end{pmatrix}$

They intersect at $(4, 6, 5)$

Solve 2 of the 3 equations, sub into the 3rd: LHS = RHS ⇒ intersect; LHS ≠ RHS ⇒ skew.

Line – Plane $l: \mathbf{r}=\mathbf{a}+\lambda\mathbf{b},\ p: \mathbf{r}\cdot\mathbf{n}=d$

	Parallel, not on the plane	Line lies in the plane	Intersects at one point
Diagram	 $l: \mathbf{r}=\mathbf{a}+\lambda\mathbf{b},\ \lambda\in\mathbb{R}$ $p: \mathbf{r}\cdot\mathbf{n}=d$	 $l: \mathbf{r}=\mathbf{a}+\lambda\mathbf{b},\ \lambda\in\mathbb{R}$ $p: \mathbf{r}\cdot\mathbf{n}=d$	 $l: \mathbf{r}=\mathbf{a}+\lambda\mathbf{b},\ \lambda\in\mathbb{R}$ $p: \mathbf{r}\cdot\mathbf{n}=d$
Condition	• normal ⊥ line: $\mathbf{b}\cdot\mathbf{n}=0$ • A does not lie on p: $\mathbf{a}\cdot\mathbf{n}\neq d$	• normal ⊥ line: $\mathbf{b}\cdot\mathbf{n}=0$ • A lies on p: $\mathbf{a}\cdot\mathbf{n}=d$	• normal not ⊥ line: $\mathbf{b}\cdot\mathbf{n}\neq 0$



Example 15.8

Find the point of intersection between the line l and plane p.

$$l:\ \mathbf{r}=\begin{pmatrix}1\\3\\1\end{pmatrix}+\lambda\begin{pmatrix}1\\-2\\1\end{pmatrix},\quad \lambda\in\mathbb{R}$$

$$p:\ \mathbf{r}\cdot\begin{pmatrix}1\\2\\1\end{pmatrix}=5$$

$$\mathbf{r}=\begin{pmatrix}1\\3\\1\end{pmatrix}+\lambda\begin{pmatrix}1\\-2\\1\end{pmatrix}=\begin{pmatrix}1+\lambda\\3-2\lambda\\1+\lambda\end{pmatrix},\quad \lambda\in\mathbb{R}$$

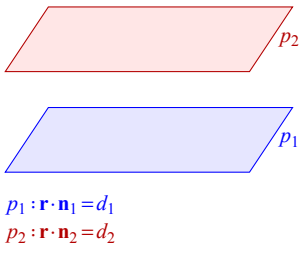
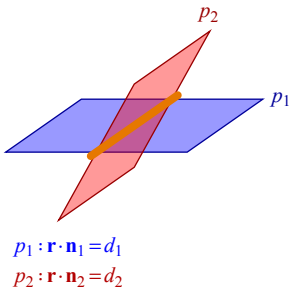
$$\begin{pmatrix}1+\lambda\\3-2\lambda\\1+\lambda\end{pmatrix}\cdot\begin{pmatrix}1\\2\\1\end{pmatrix}=5$$

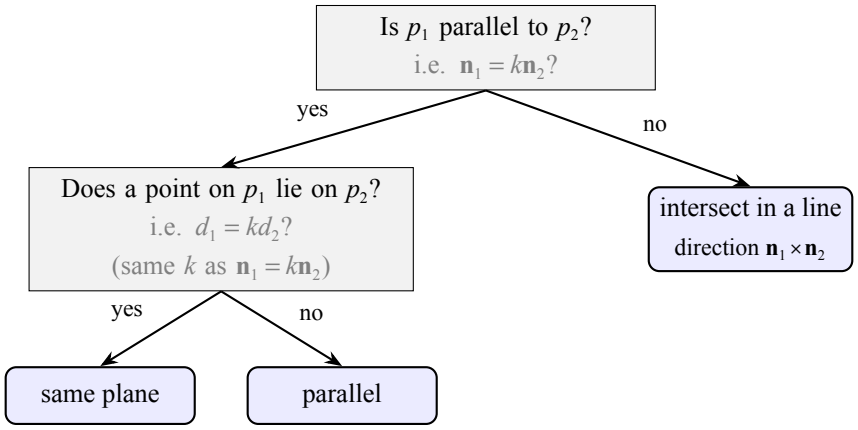
$$8-2\lambda=5\ \Rightarrow\ \lambda=\tfrac{3}{2}$$

$$\therefore\ (2.5,0,2.5)$$

Plane – Plane $p_1 : \mathbf{r}\cdot\mathbf{n}_1 = d_1,\ p_2 : \mathbf{r}\cdot\mathbf{n}_2 = d_2$

Let $p_1 :\ \mathbf{r}\cdot\mathbf{n}_1 = d_1$ and $p_2 :\ \mathbf{r}\cdot\mathbf{n}_2 = d_2$ be distinct planes.

	Parallel	Intersecting in a line
Diagram		
Condition	$\mathbf{n}_1 \parallel \mathbf{n}_2, p_1 \neq p_2$	$\mathbf{n}_1 \nparallel \mathbf{n}_2$



Example 15.9

Find the intersection between the planes:

$$3x+2y+z=10$$
$$x-2y+z=5$$

GC: $x=\dfrac{15}{4}-\dfrac{1}{2}z$

$y=-\dfrac{5}{8}+\dfrac{1}{4}z$

$z=z$

$$\therefore l:\ \mathbf{r}=\begin{pmatrix}\frac{15}{4}\\-\frac{5}{8}\\0\end{pmatrix}+\lambda\begin{pmatrix}-\frac{1}{2}\\\frac{1}{4}\\1\end{pmatrix},\quad \lambda\in\mathbb{R}$$

Example 15.10

Sometimes, we may not be able to use our GC to find the line of intersection between two planes. The steps to find it manually are:

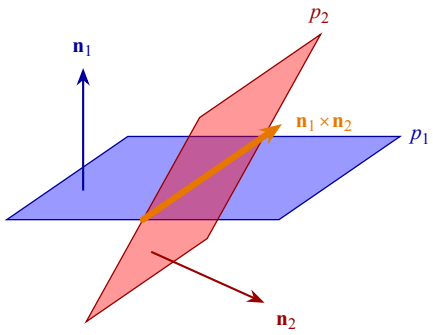
$$p_1: \mathbf{r}\cdot\begin{pmatrix}1\\1\\1\end{pmatrix}=6, \qquad p_2: \mathbf{r}\cdot\begin{pmatrix}1\\-1\\2\end{pmatrix}=2$$

Step 1: direction $=\mathbf{n}_1\times\mathbf{n}_2$

$$\mathbf{b}=\begin{pmatrix}1\\1\\1\end{pmatrix}\times\begin{pmatrix}1\\-1\\2\end{pmatrix}$$

$$=\begin{pmatrix}1(2)-1(-1)\\1(1)-1(2)\\1(-1)-1(1)\end{pmatrix}$$

$$=\begin{pmatrix}3\\-1\\-2\end{pmatrix}$$



Step 2: find a point that lies on both planes

$$p_1: x+y+z=6$$

$$p_2: x-y+2z=2$$

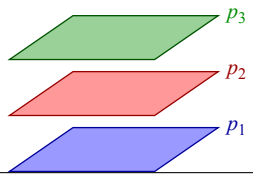
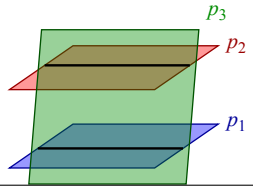
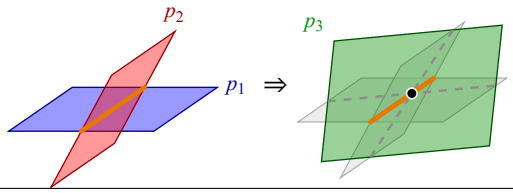
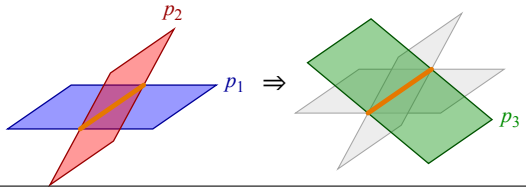
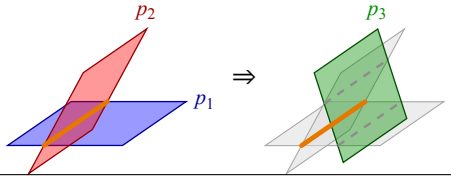
$$\text{Let } z=0$$

$$\left.\begin{matrix}x+y=6\\x-y=2\end{matrix}\right\}\Rightarrow\begin{matrix}x=4\\y=2\end{matrix}$$

$$\therefore l: \mathbf{r}=\begin{pmatrix}4\\2\\0\end{pmatrix}+\lambda\begin{pmatrix}3\\-1\\-2\end{pmatrix}, \quad \lambda\in\mathbb{R}$$

15.7 Interaction between 3 Planes

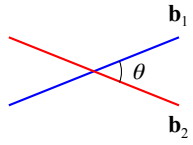
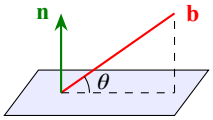
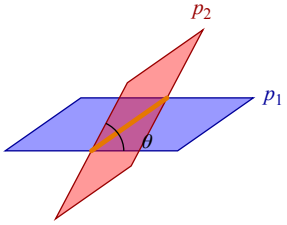
Let $p_1: \mathbf{r}\cdot\mathbf{n}_1 = d_1$, $p_2: \mathbf{r}\cdot\mathbf{n}_2 = d_2$ and $p_3: \mathbf{r}\cdot\mathbf{n}_3 = d_3$ be distinct planes.

Case	Diagram(s)	Condition l: line of intersection of p_1 and p_2 ($\mathbf{n}_1 \nparallel \mathbf{n}_2$)
1. All parallel		$\mathbf{n}_1 \parallel \mathbf{n}_2 \parallel \mathbf{n}_3$
2. Two parallel, one not		$\mathbf{n}_1 \parallel \mathbf{n}_2$, but $\mathbf{n}_3$ not parallel to them
3. All intersect at 1 point (most common)		$l: \mathbf{r}=\mathbf{a}+\lambda\mathbf{b},\ \lambda\in\mathbb{R}$ intersects p_3 (see p. 78)
4. All intersect in a line		$l: \mathbf{r}=\mathbf{a}+\lambda\mathbf{b},\ \lambda\in\mathbb{R}$ lies on p_3 (see p. 78)
5. Form a triangular prism (no single point of intersection)		$l: \mathbf{r}=\mathbf{a}+\lambda\mathbf{b},\ \lambda\in\mathbb{R}$ • $l \parallel p_3$: $\mathbf{b}\cdot\mathbf{n}_3 = 0$ • $l \not\subset p_3$: $\mathbf{a}\cdot\mathbf{n}_3 \neq d_3$ (see p. 78)

Note

To analyse the interaction between 3 arbitrary planes, we can study how the **line of intersection between p_1 and p_2** interacts with p_3.

15.8 Angles

Line – Line	Line – Plane	Plane – Plane
		
$\cos\theta = \dfrac{\lvert\mathbf{b}_1\cdot\mathbf{b}_2\rvert}{\lvert\mathbf{b}_1\rvert\lvert\mathbf{b}_2\rvert}$	$\sin\theta = \left\lvert\dfrac{\mathbf{n}\cdot\mathbf{b}}{\lvert\mathbf{n}\rvert\lvert\mathbf{b}\rvert}\right\rvert$	$\cos\theta = \left\lvert\dfrac{\mathbf{n}_1\cdot\mathbf{n}_2}{\lvert\mathbf{n}_1\rvert\lvert\mathbf{n}_2\rvert}\right\rvert$
directions	sin: $\angle(\mathbf{n},\mathbf{b}) = 90^\circ - \theta$	normals

θ = acute angle.

Each formula has a modulus in the numerator $\because \theta$ must be acute.

Example 15.11

Find the angle between the planes $3x-y+2z=3$ and $y=2$.

$$\cos\theta = \frac{|-1|}{\sqrt{14}\cdot 1}$$
$$\theta = 74.5^\circ$$

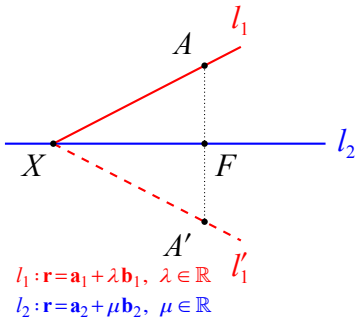
Example 15.12

Find the angle between the line l and plane p.

$$l:\ \mathbf{r} = \begin{pmatrix}-2\\-4\\5\end{pmatrix} + \lambda\begin{pmatrix}3\\5\\-4\end{pmatrix},\quad \lambda\in\mathbb{R}$$
$$p:\ \mathbf{r}\cdot\begin{pmatrix}0\\1\\0\end{pmatrix} = 1$$
$$\sin\theta = \frac{5}{\sqrt{50}\cdot 1} = \frac{1}{\sqrt{2}}$$
$$\theta = 45^\circ$$

15.9 Reflection of a Line

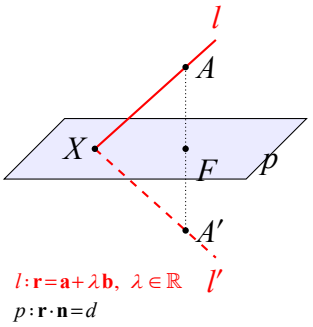
Line l_1 in line l_2 (intersect at X)



1. $X=l_1\cap l_2$
2. $A\in l_1$: foot F on l_2
3. $\overrightarrow{OF}=\dfrac{\overrightarrow{OA}+\overrightarrow{OA'}}{2}$ (midpoint/ratio theorem)
4. l_1': $\mathbf{r}=\overrightarrow{OX}+\mu\overrightarrow{XA'},\ \mu\in\mathbb{R}$

If $l\ //\ p$: l' has the same direction, through A'.

Line l in plane p (intersect at X)



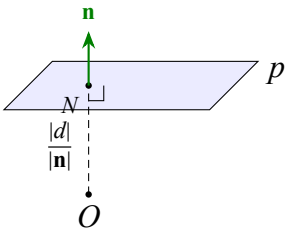
1. $X=l\cap p$
2. $A\in l$: foot F on p
3. $\overrightarrow{OF}=\dfrac{\overrightarrow{OA}+\overrightarrow{OA'}}{2}$ (midpoint/ratio theorem)
4. l': $\mathbf{r}=\overrightarrow{OX}+\mu\overrightarrow{XA'},\ \mu\in\mathbb{R}$

15.10 Distance Between Parallel Planes

$p: \mathbf{r}\cdot\mathbf{n}=d$

Distance of plane from origin

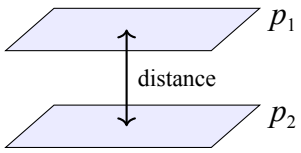
$$\text{distance from } O=\frac{|d|}{|\mathbf{n}|}$$



$p_1: \mathbf{r}\cdot\mathbf{n}=d_1,\qquad p_2: \mathbf{r}\cdot\mathbf{n}=d_2$

Distance

$$\text{distance}=\left|\frac{d_1}{|\mathbf{n}|}-\frac{d_2}{|\mathbf{n}|}\right|$$



Same $\mathbf{n}$ for both planes first. $\mathbf{r}\cdot\mathbf{n}=0$: plane through O.

Example 15.13

Find the distance between the two planes $x+2y-3z=2$ and $2x+4y-6z=-10$.

$$2x+4y-6z=-10 \Rightarrow x+2y-3z=-5$$

$$\text{distance}=\left|\frac{2}{\sqrt{14}}-\frac{-5}{\sqrt{14}}\right|=\frac{7}{\sqrt{14}}=\frac{\sqrt{14}}{2}$$